

(12) **Patent Application Publication**
MITSUNAKA

(43) **Pub. Date:** **Sep. 4, 2025**

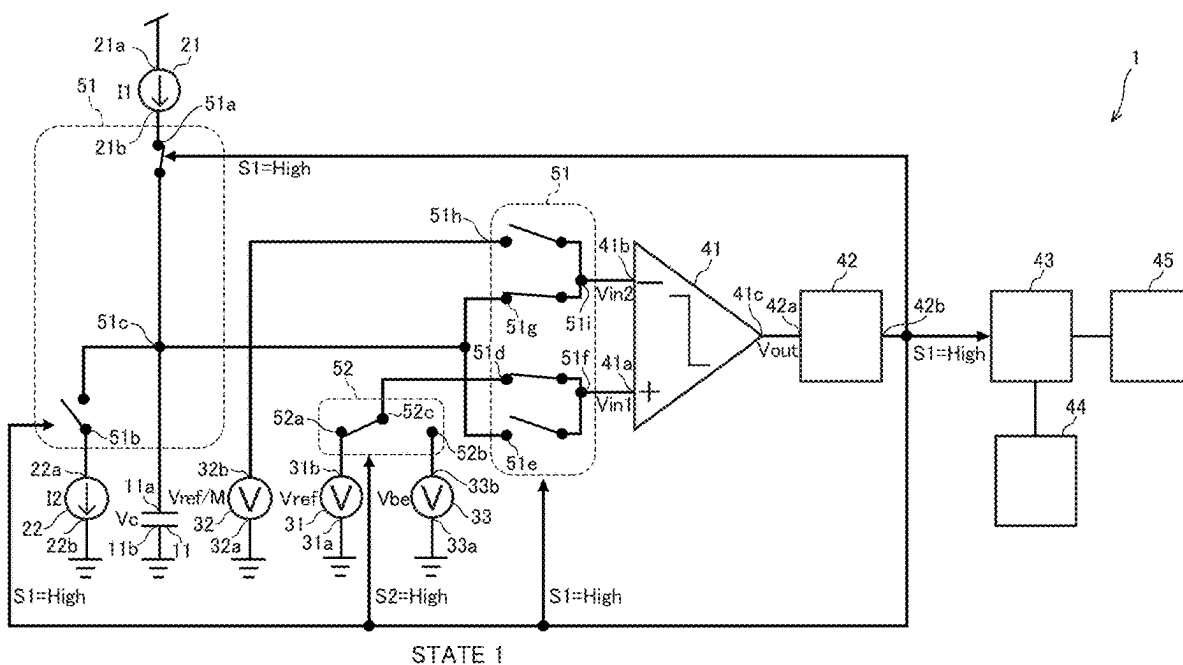


FIG. 1B

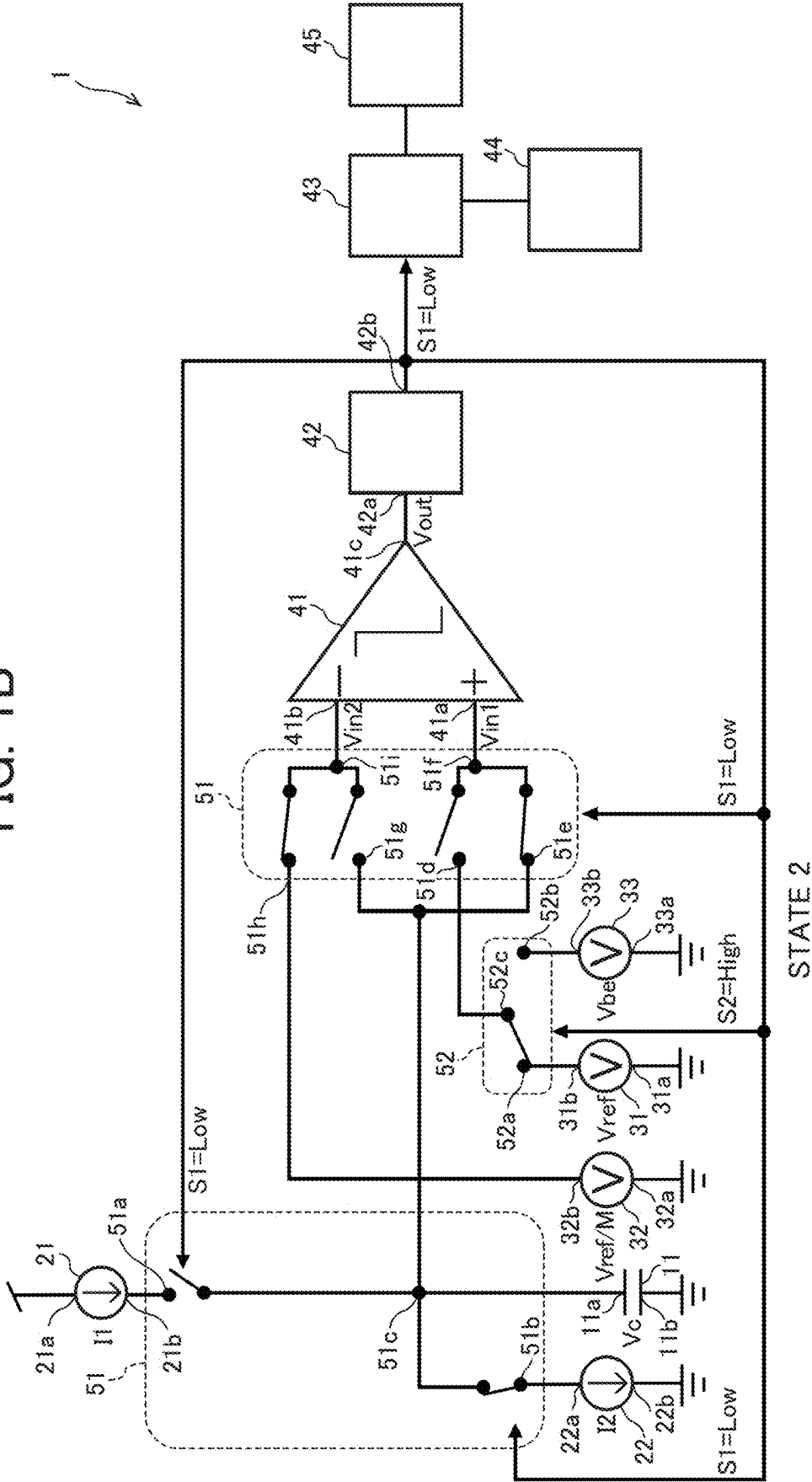


FIG. 1C

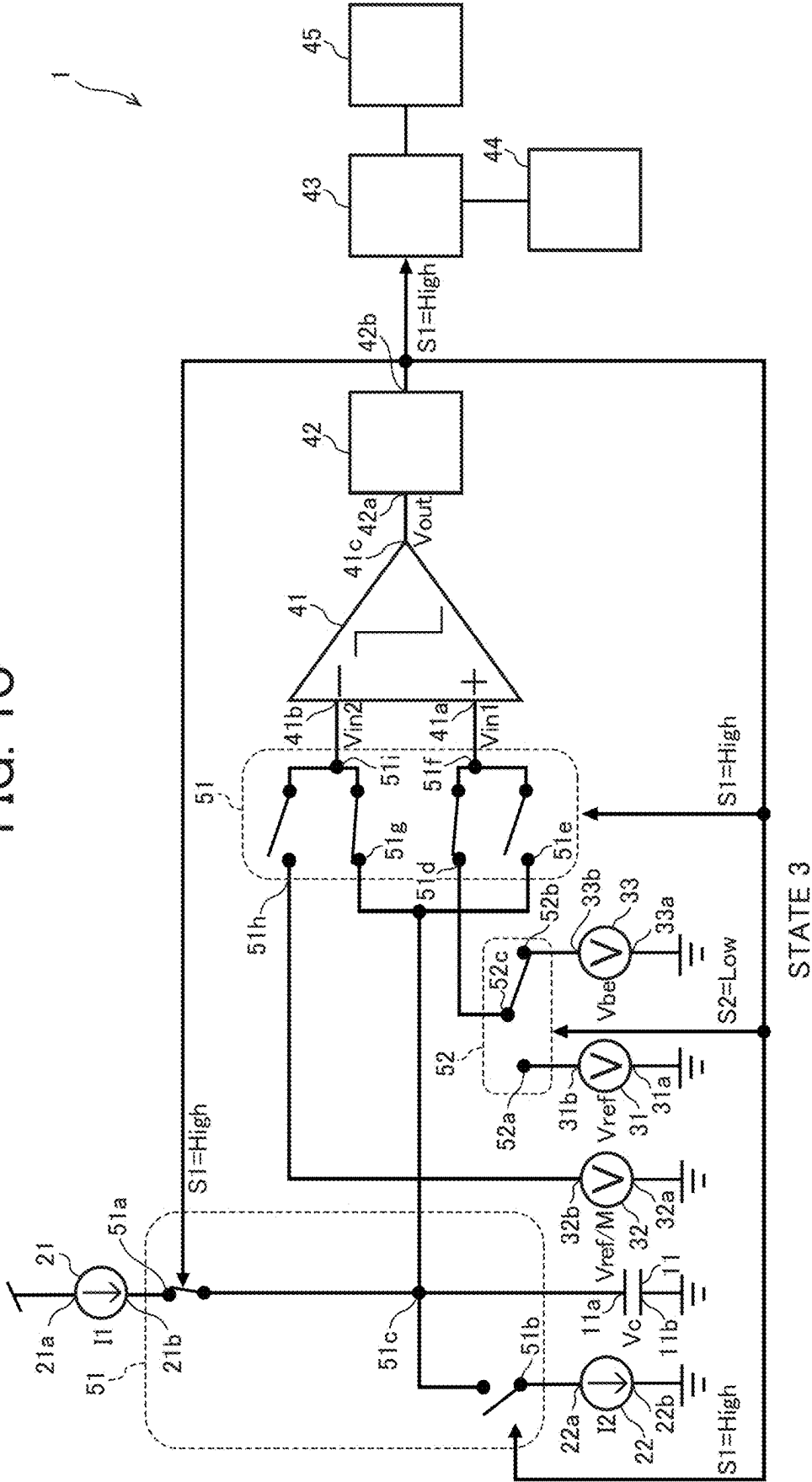


FIG. 1D

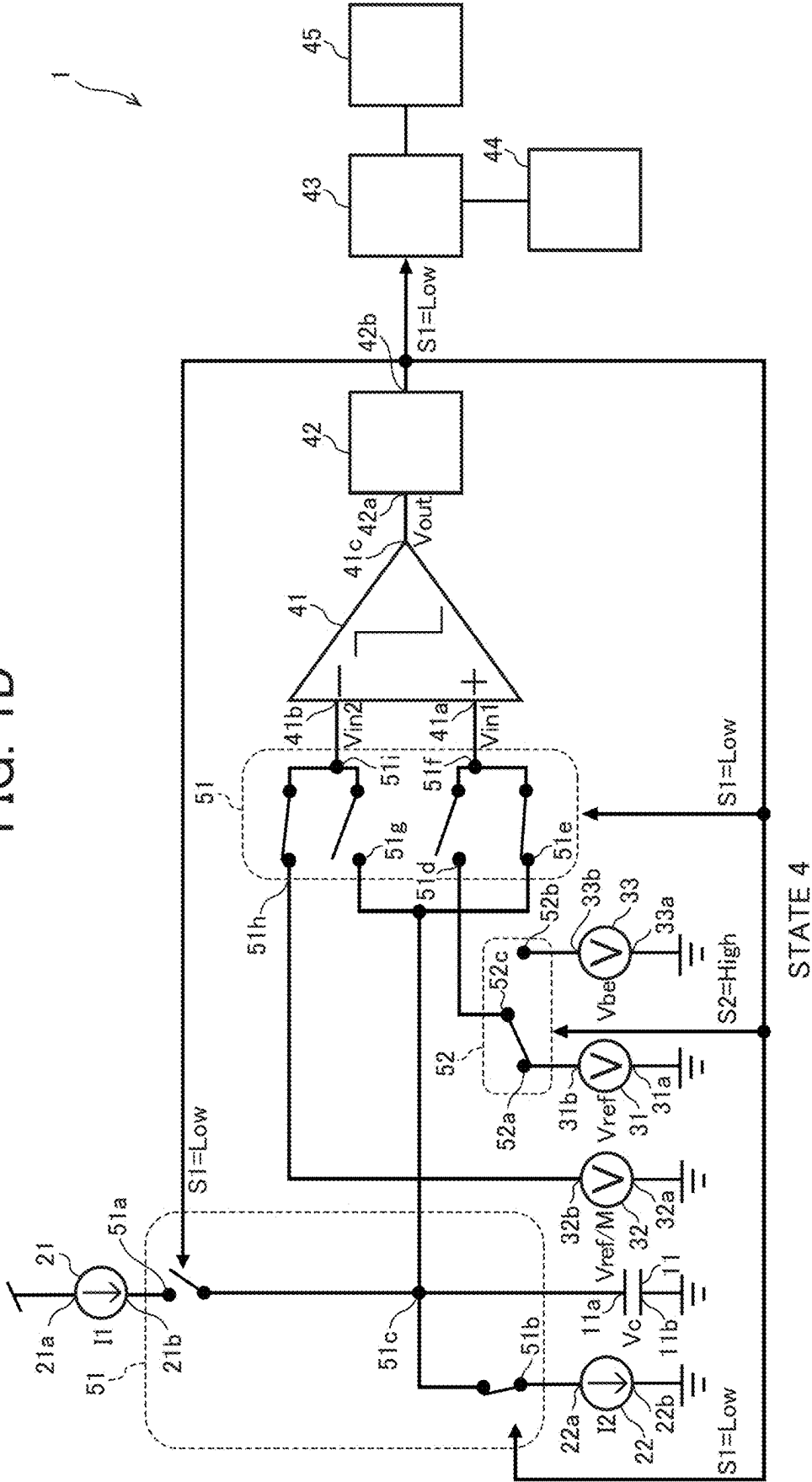


FIG. 2A

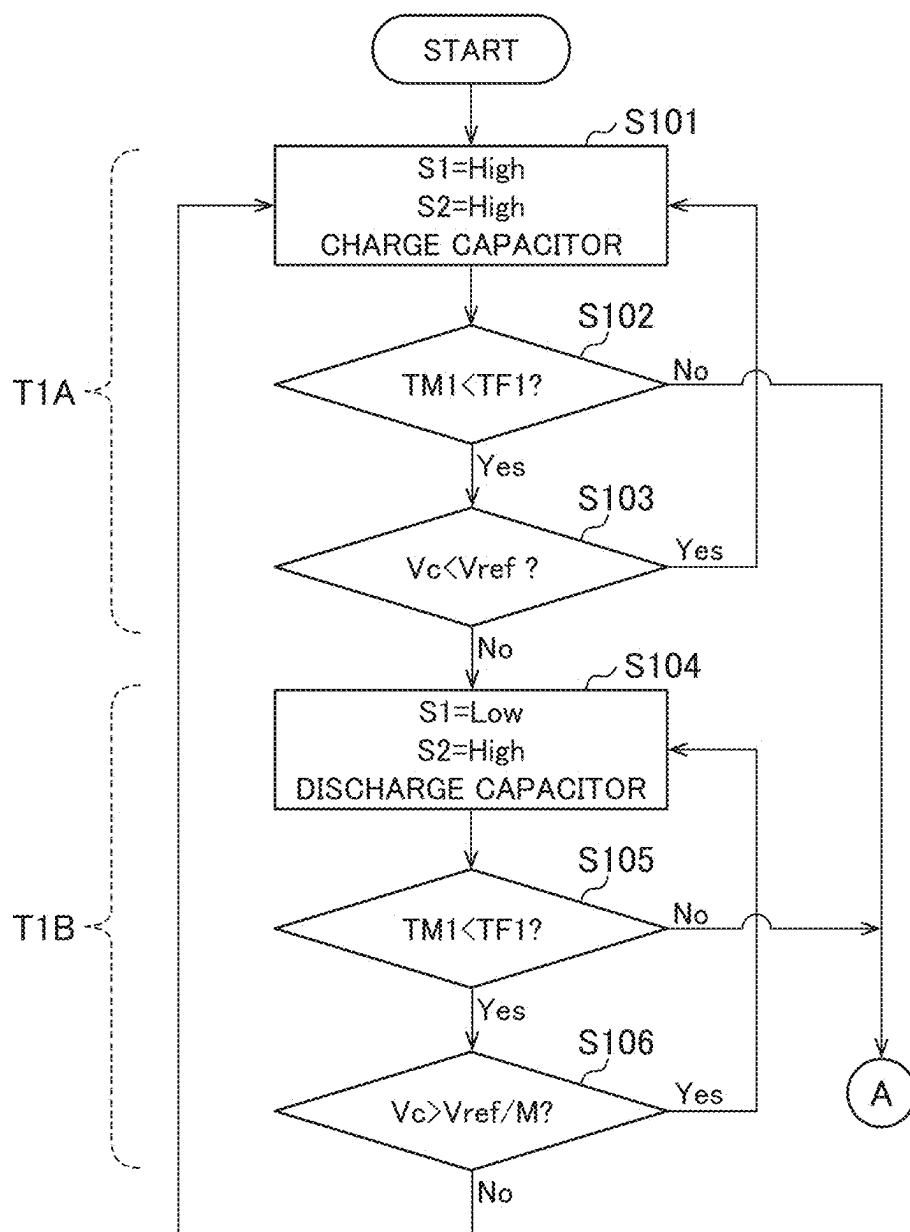


FIG. 2B

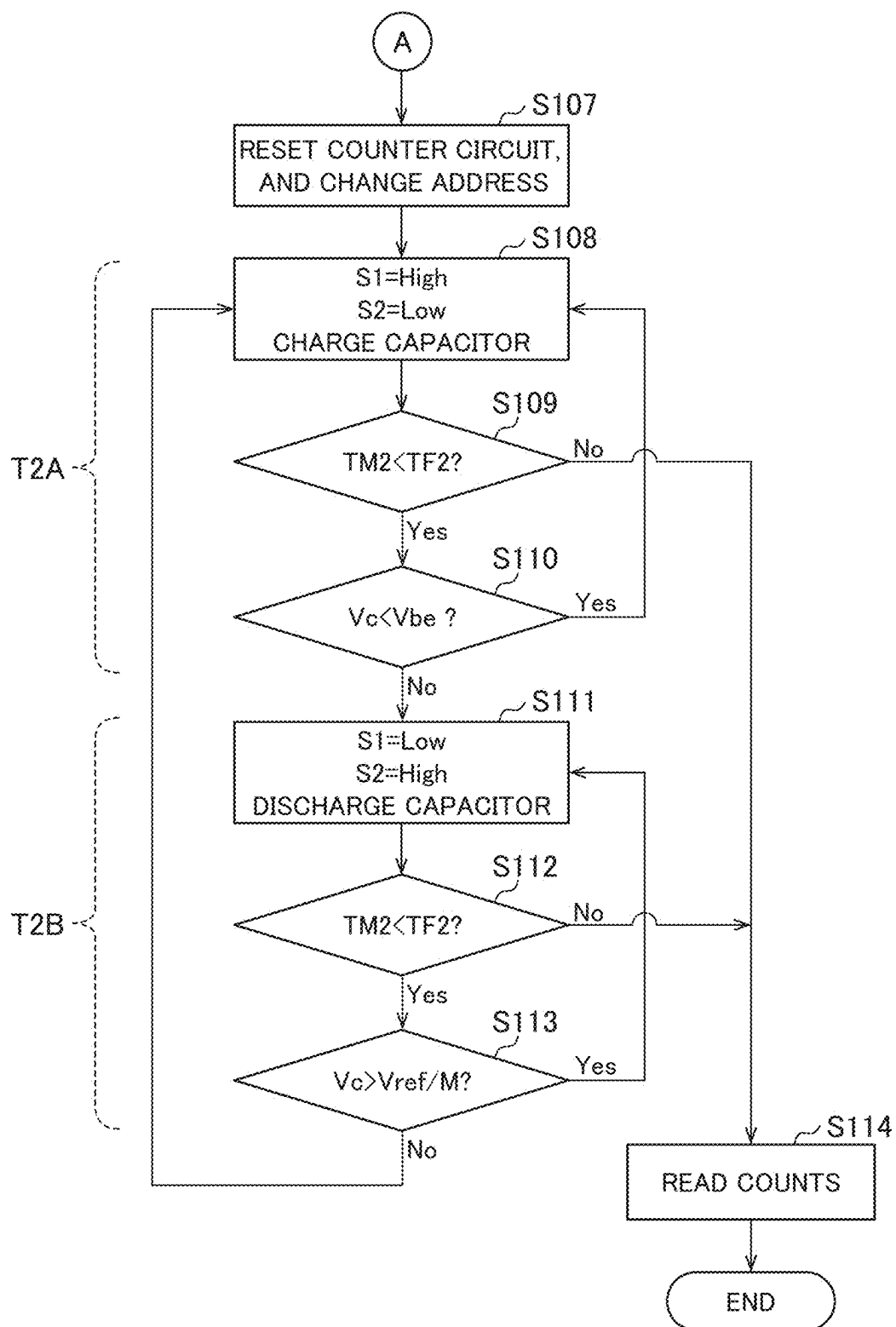
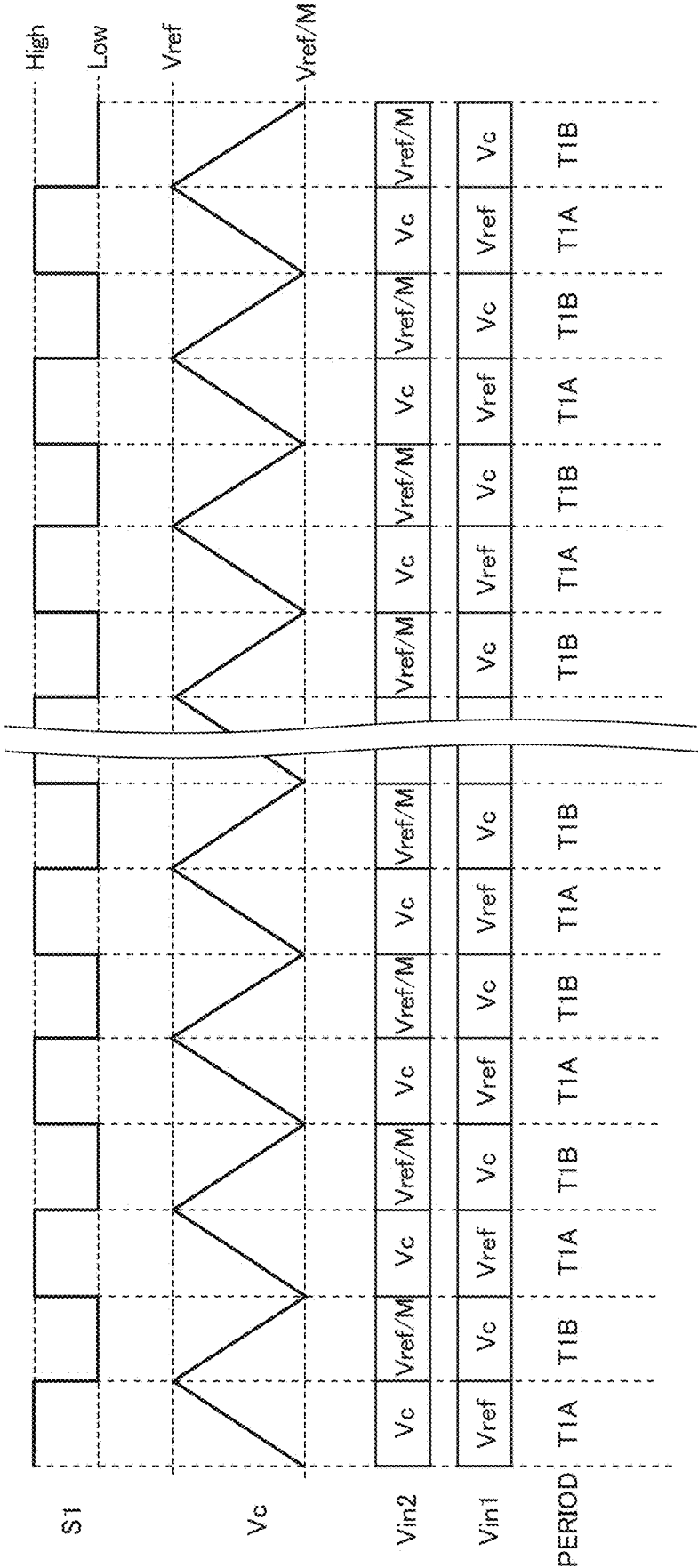


FIG. 3A



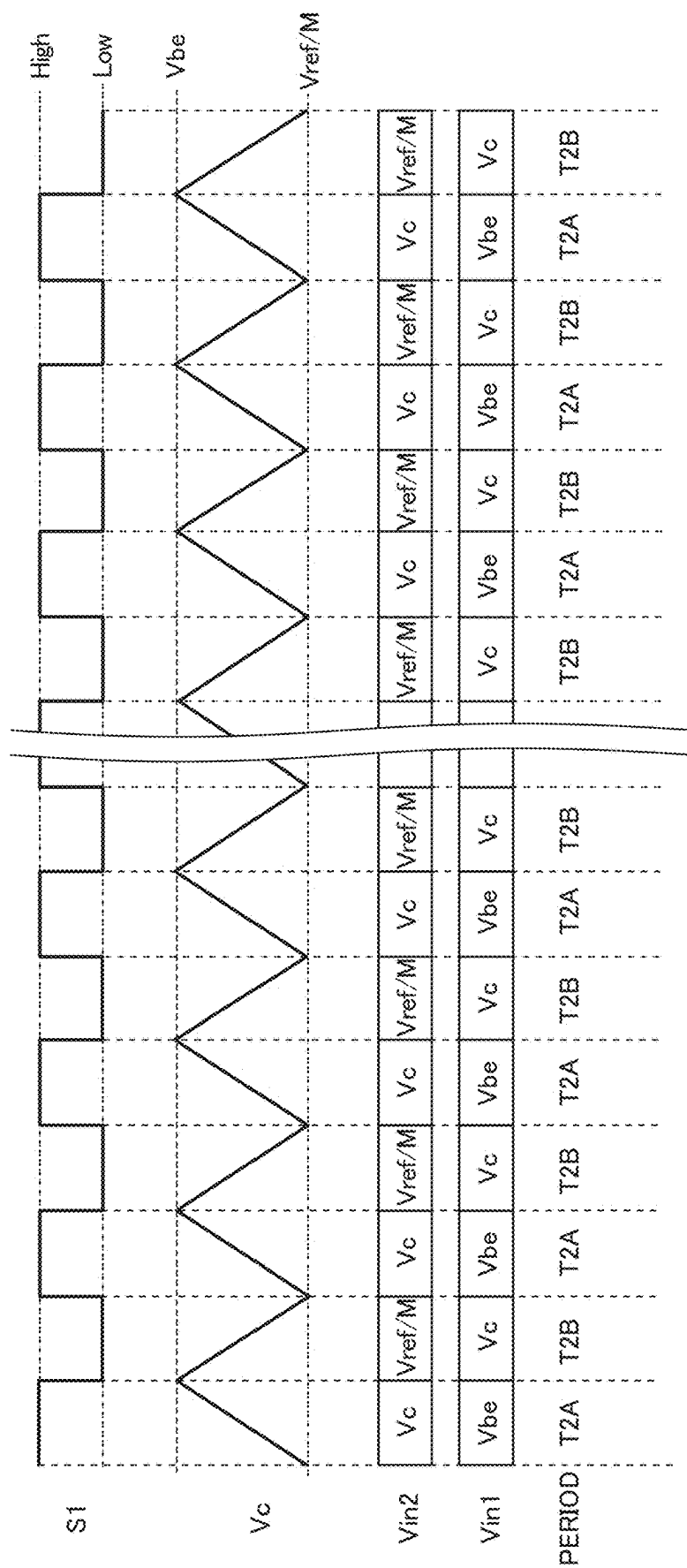


FIG. 4

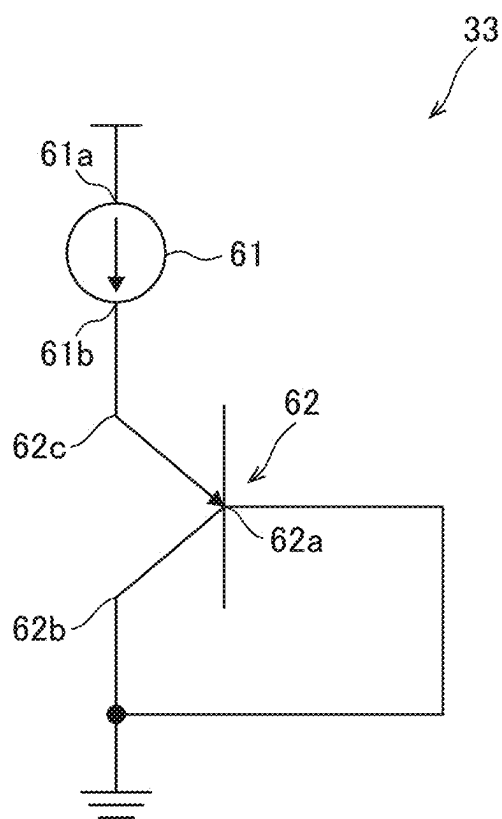
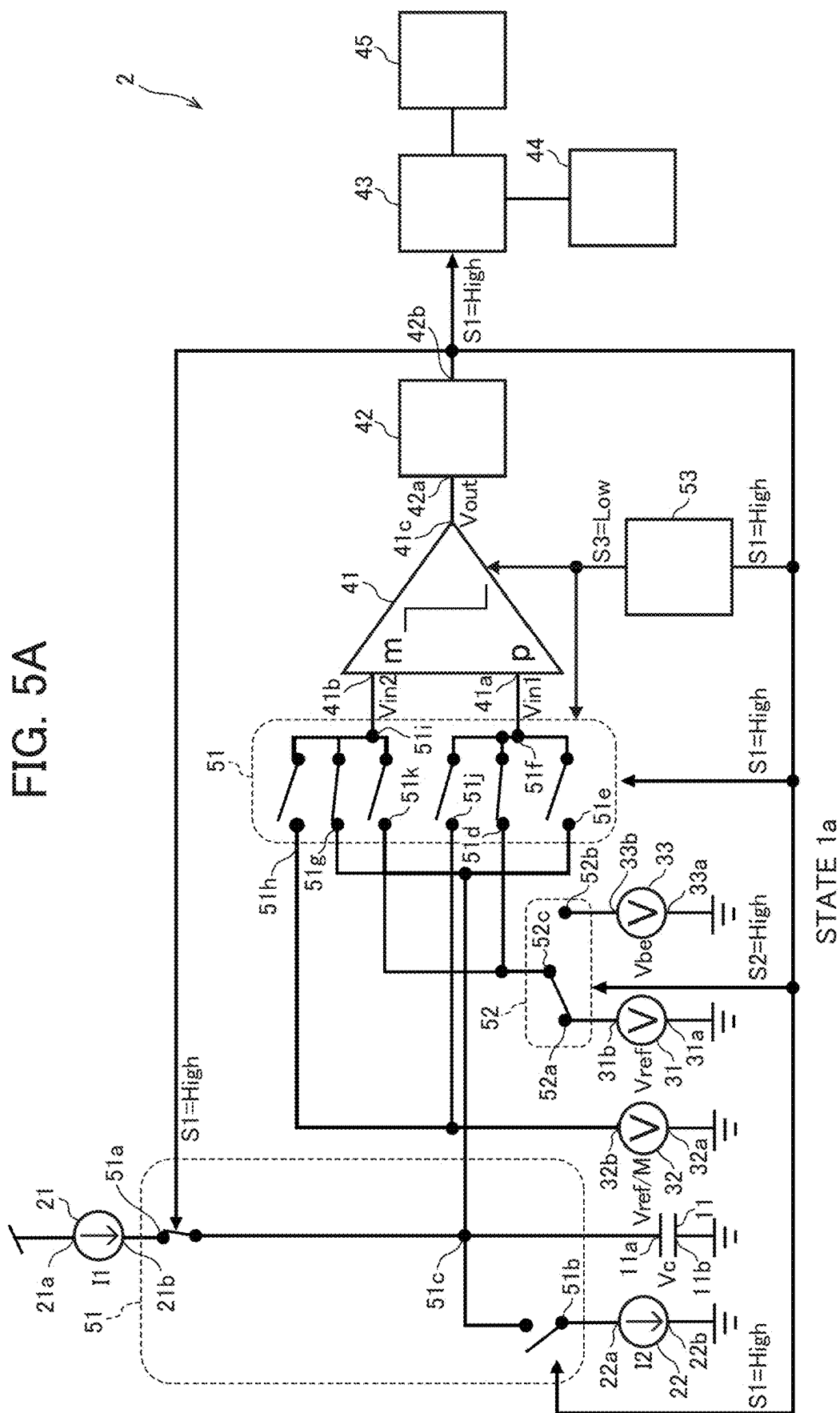
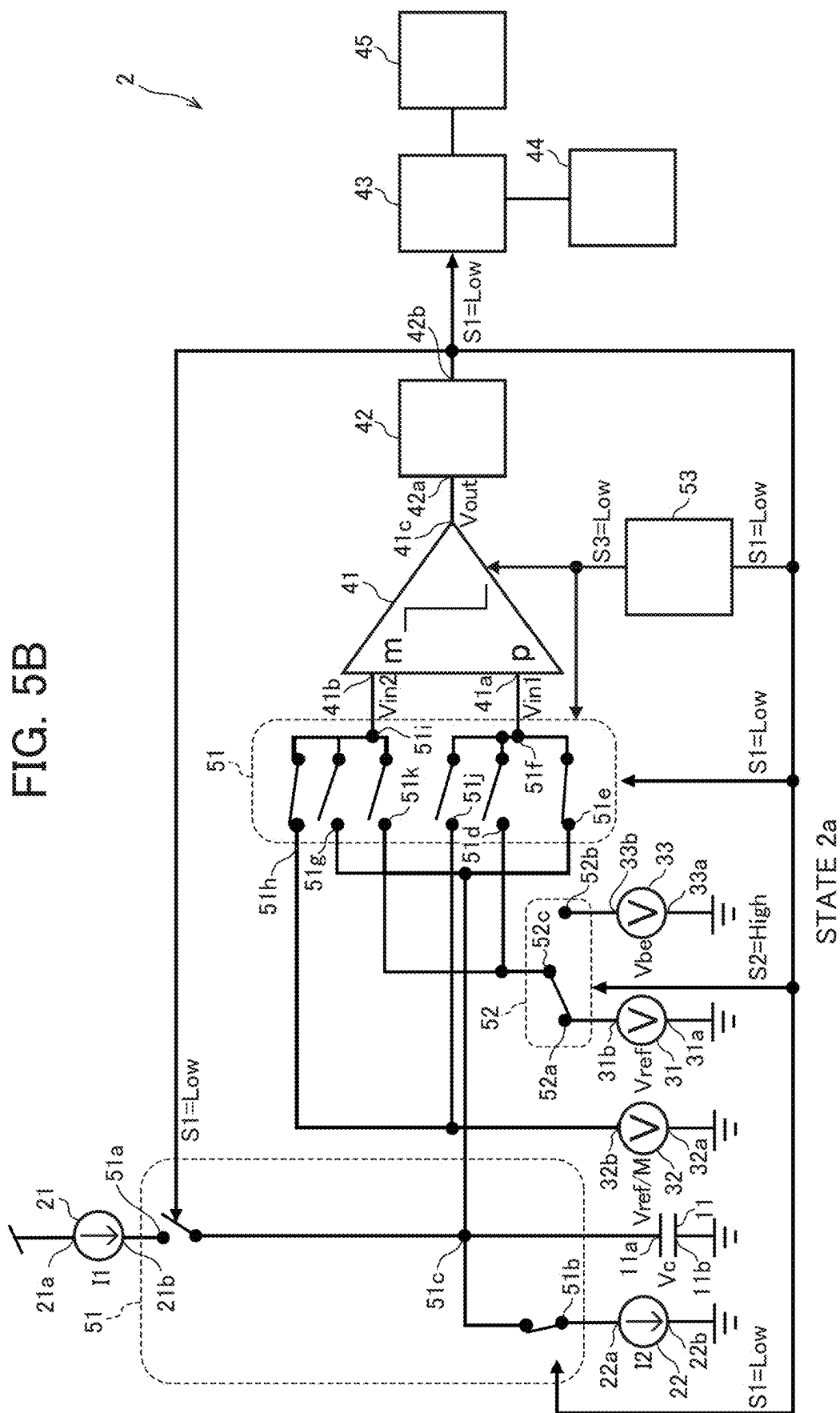


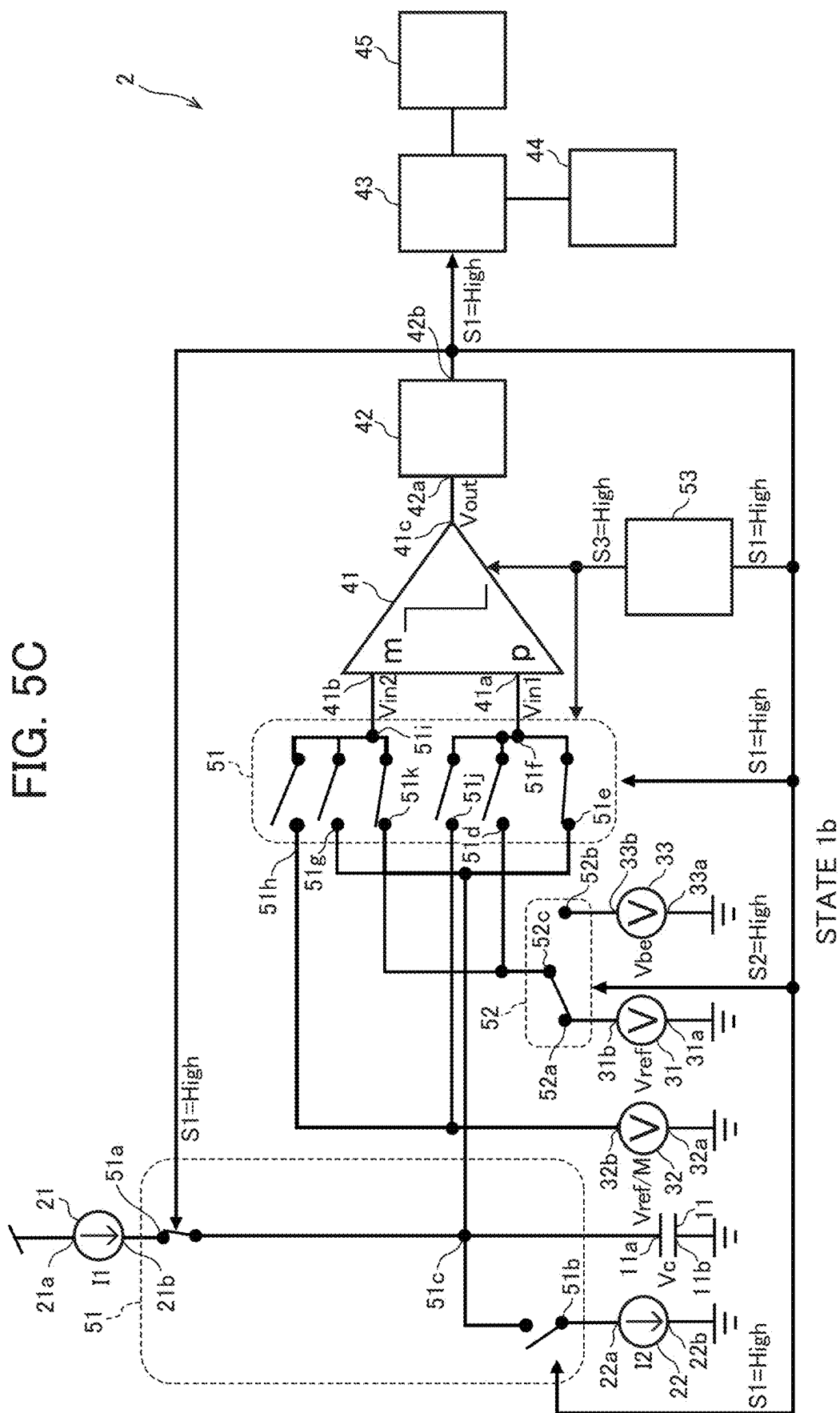
FIG. 5A



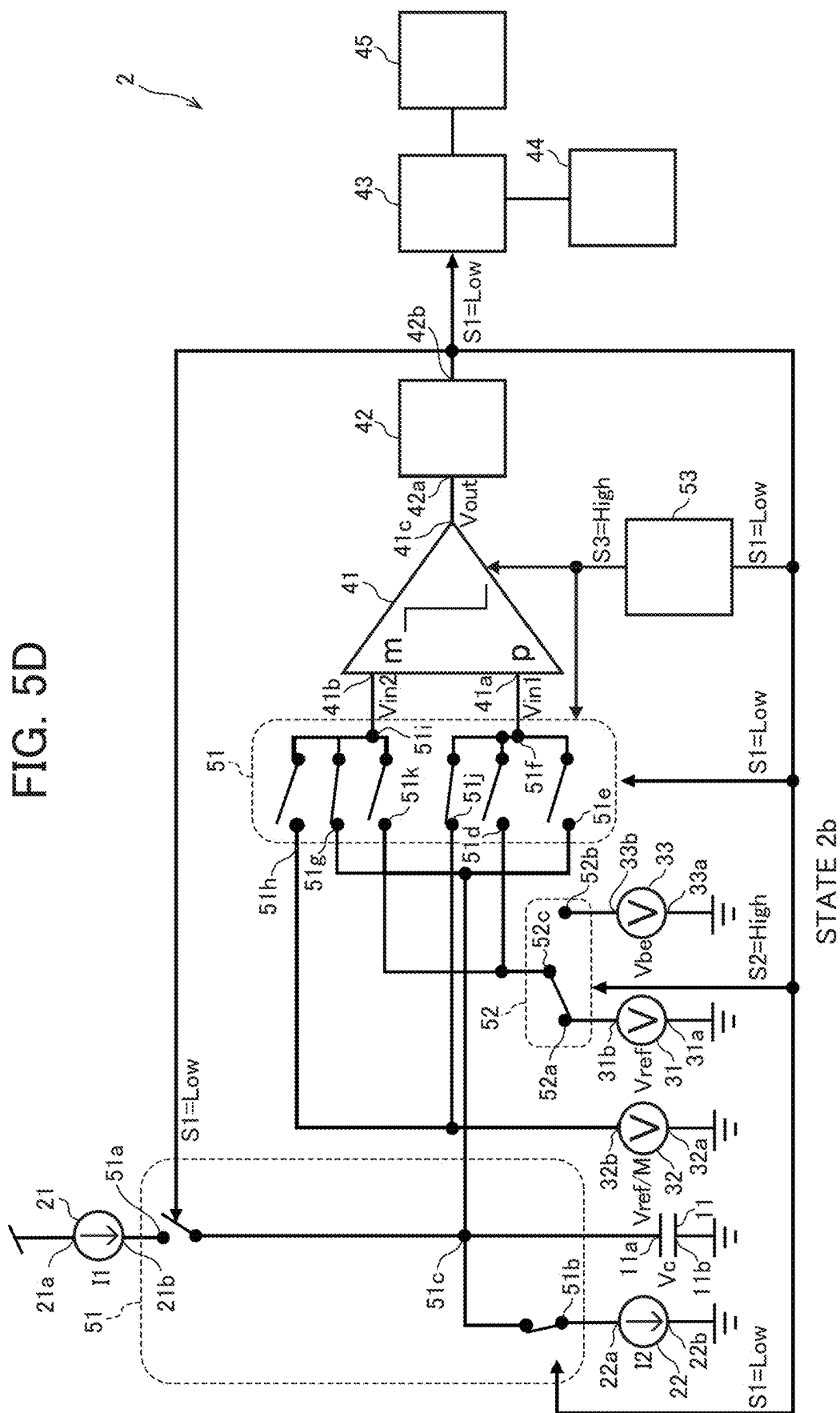
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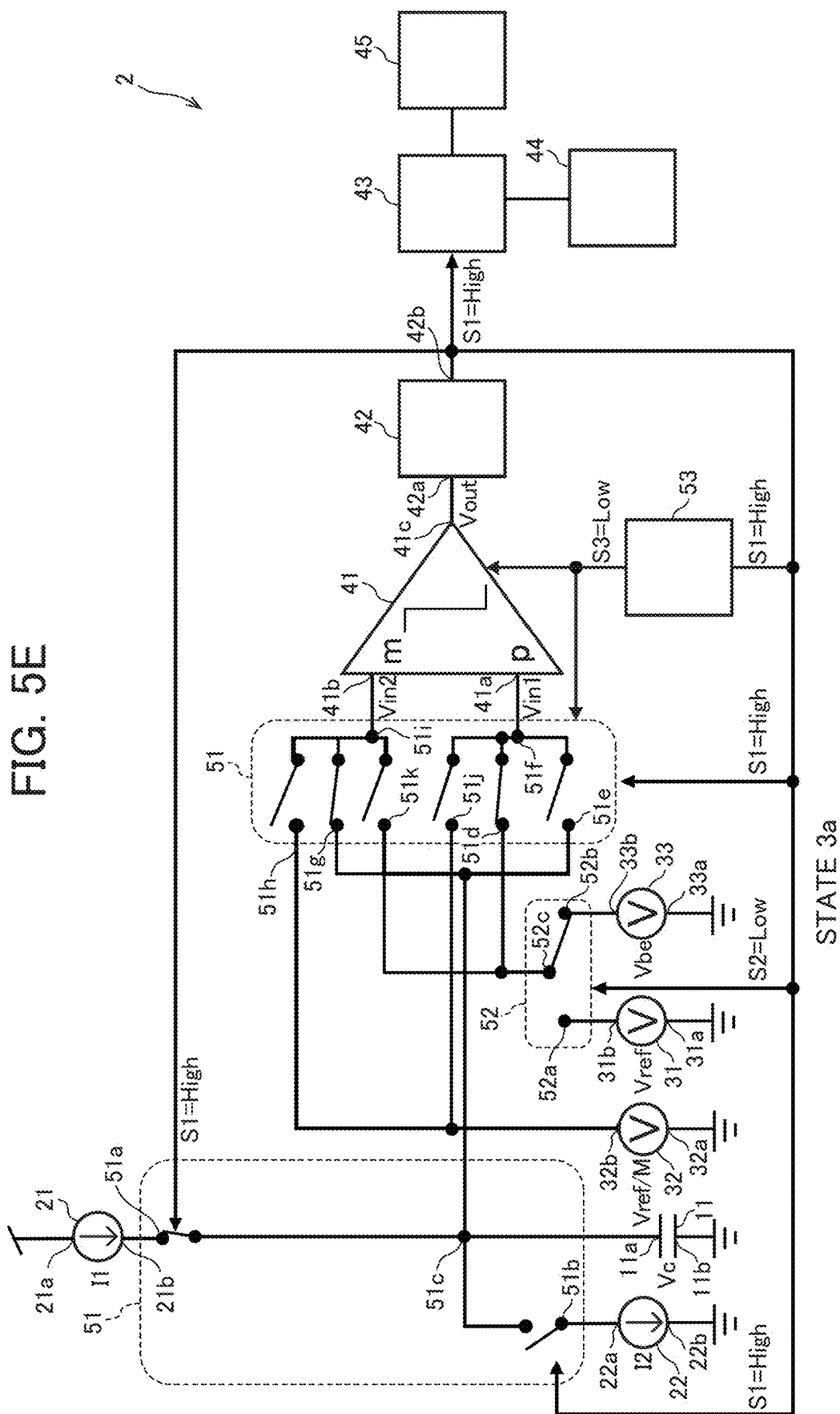


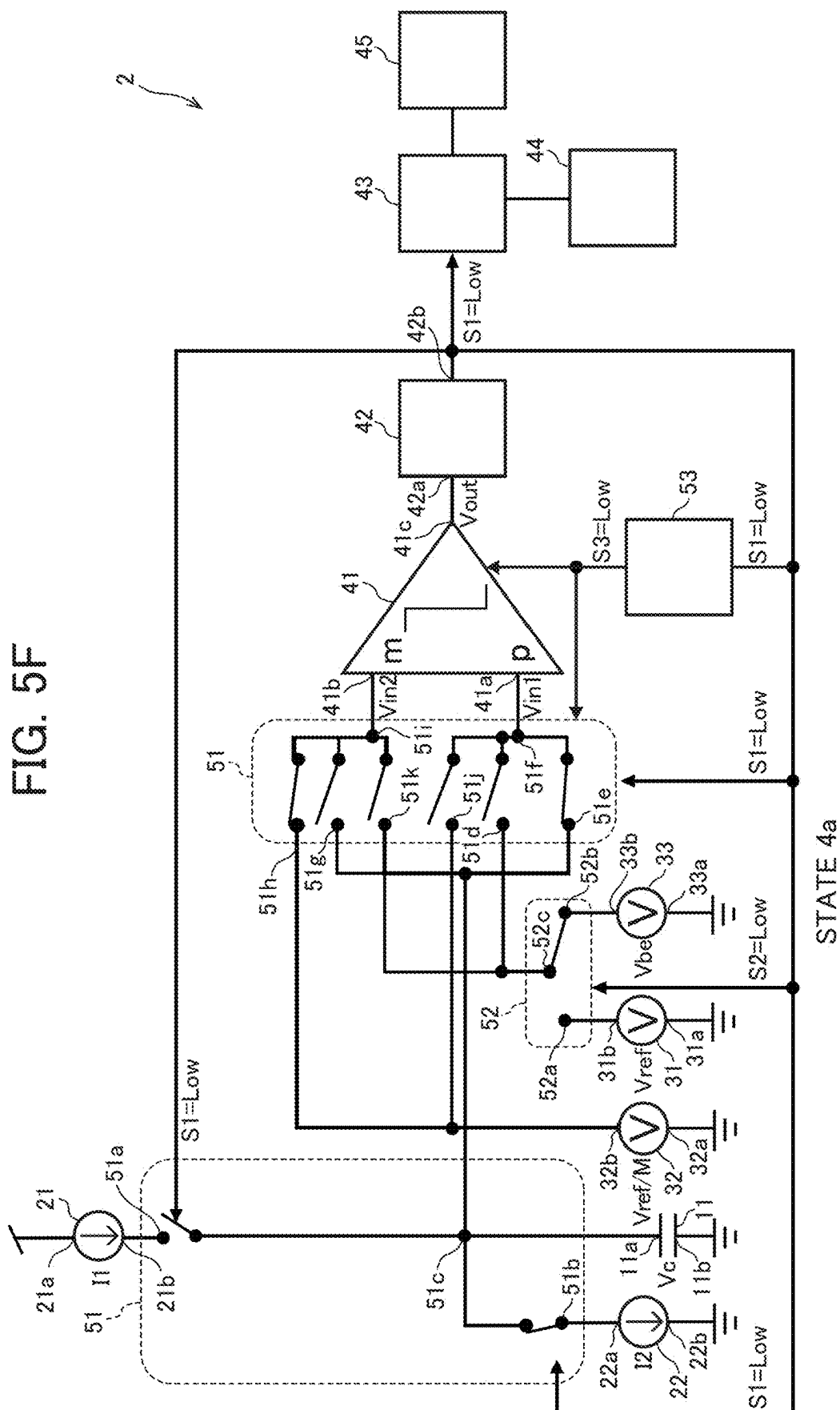
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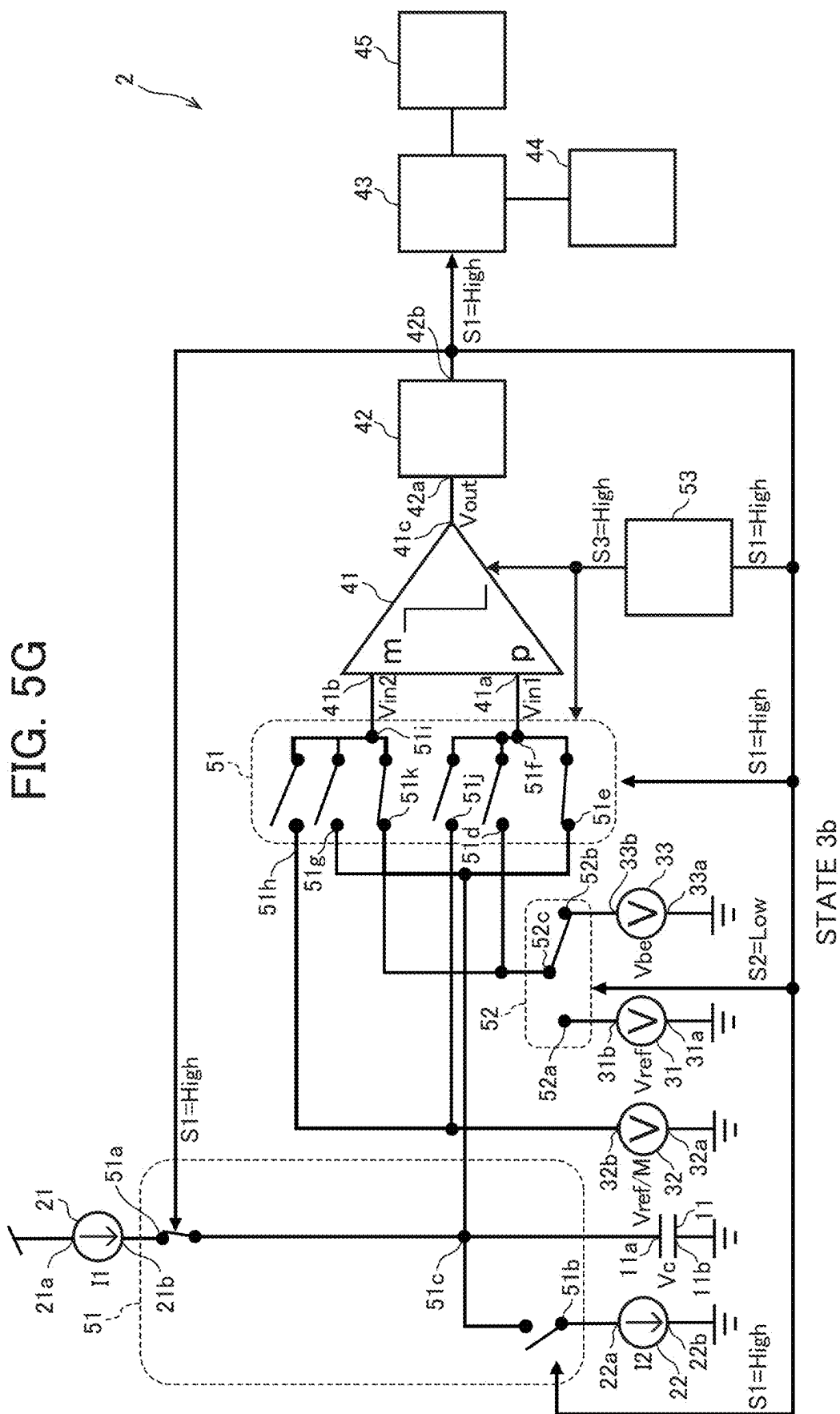


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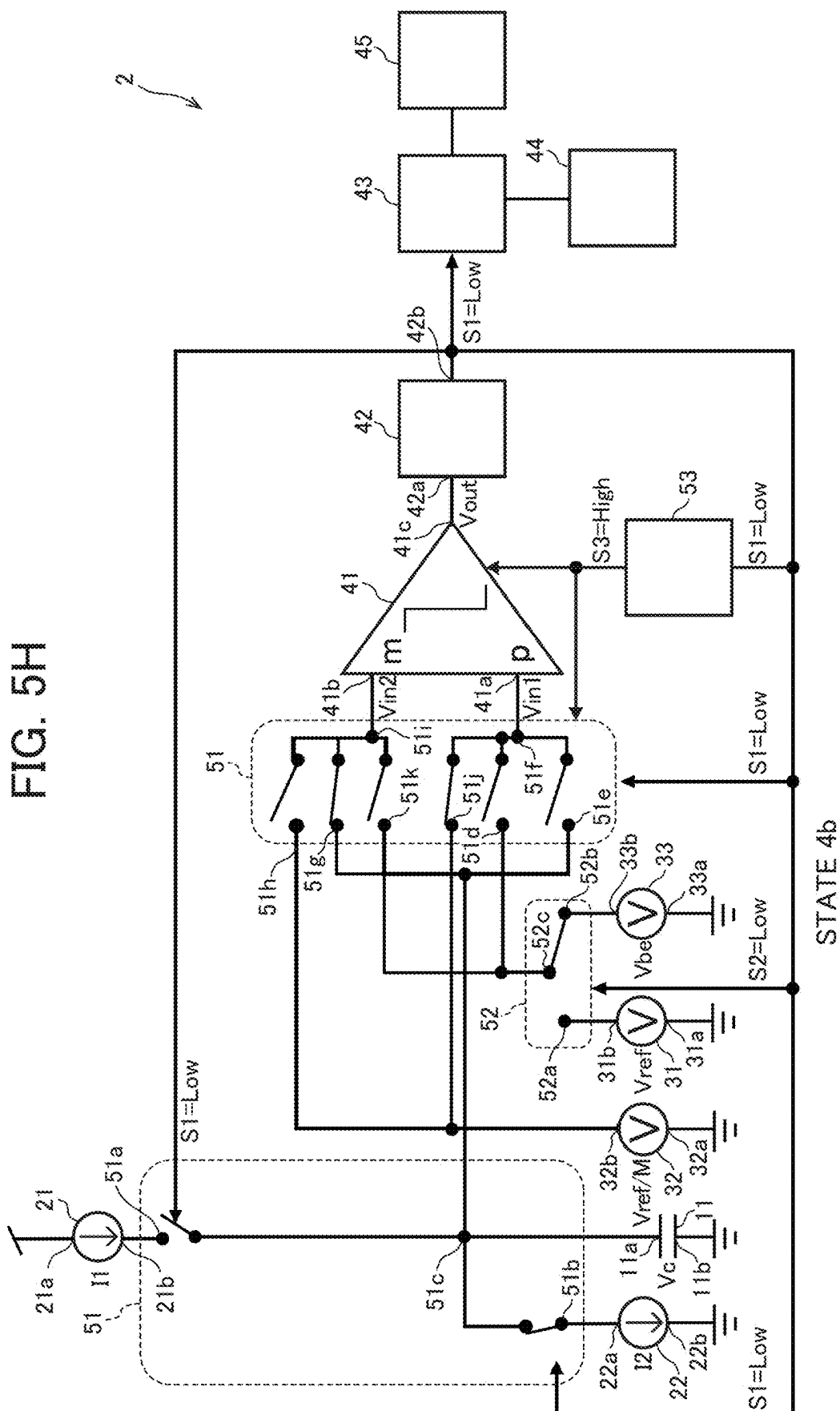


FIG. 7A

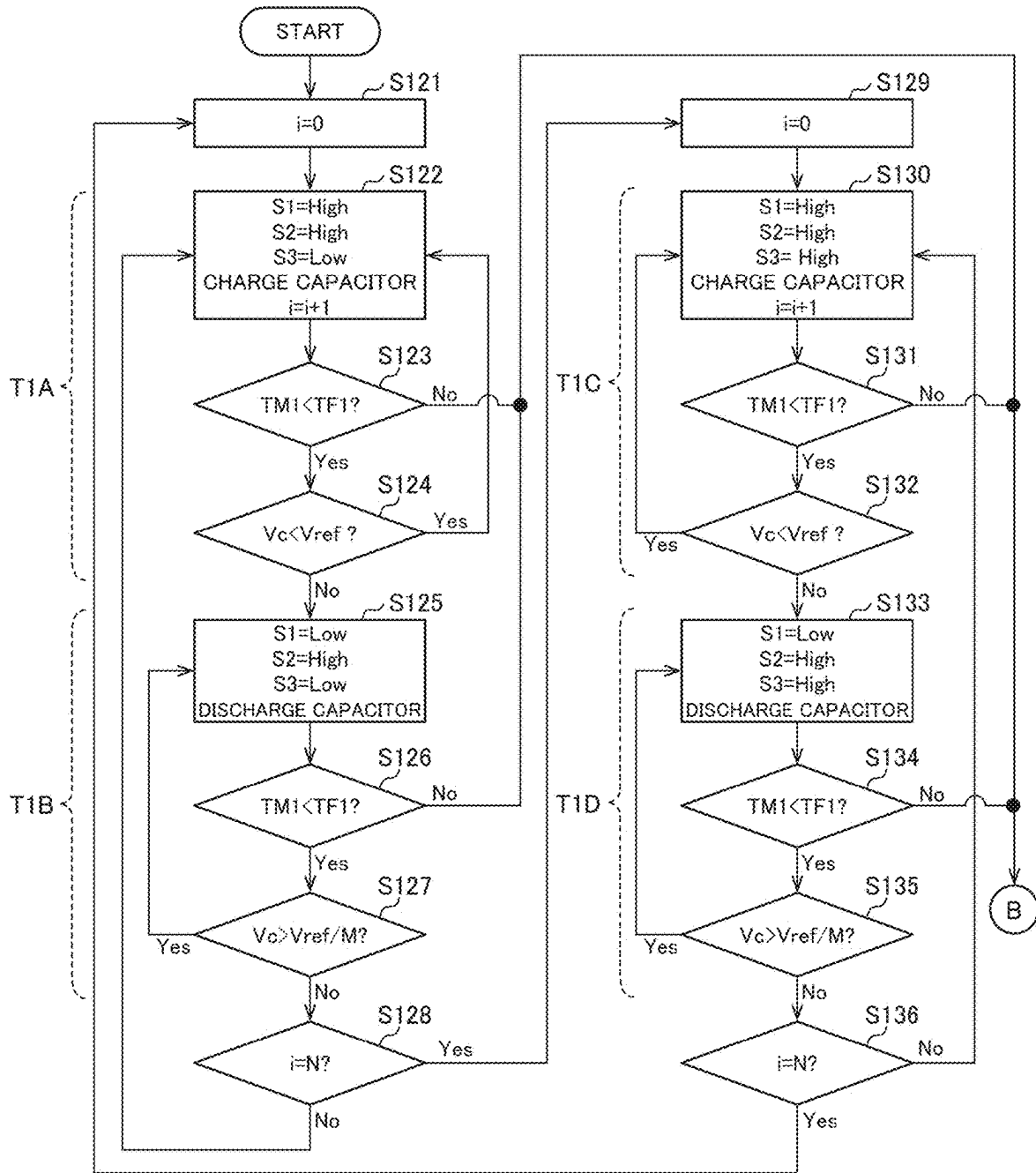
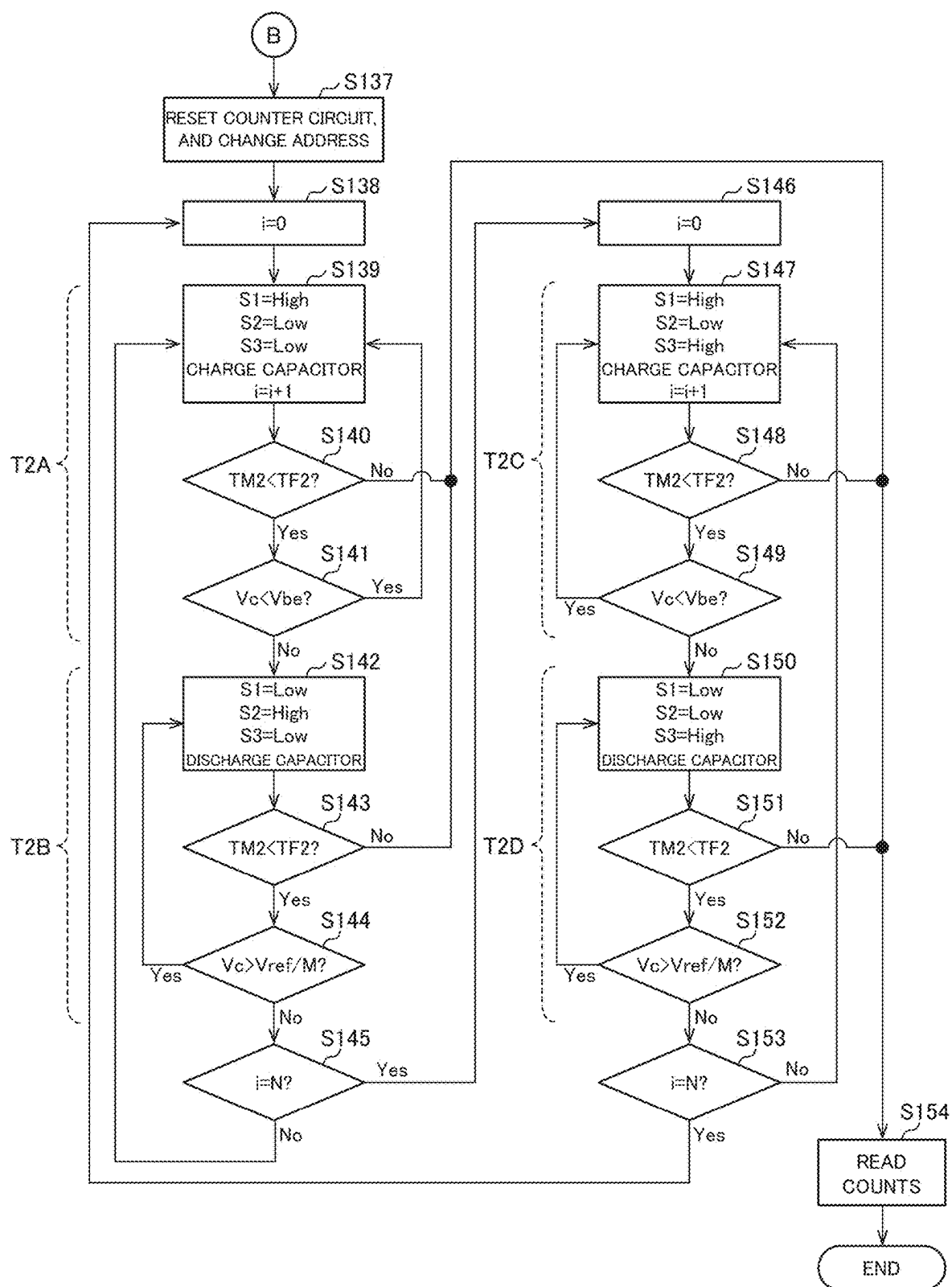
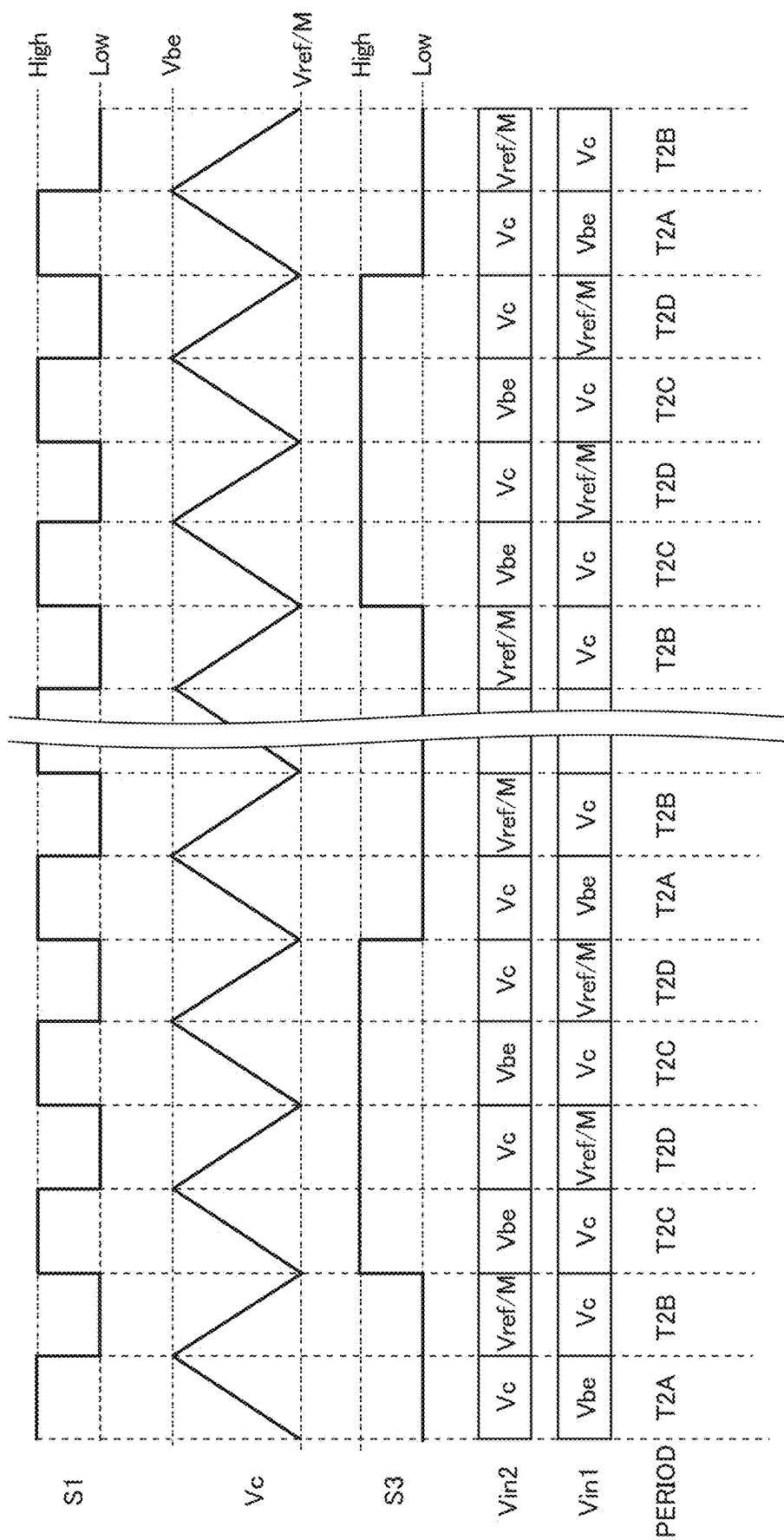


FIG. 7B





TEMPERATURE MEASURING CIRCUIT

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority from Japanese Application JP 2024-029515, the content of which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to a temperature measuring circuit.

2. Description of the Related Art

[0003] United States Unexamined Patent Application Publication No. 2022/0163402 discloses a temperature sensor.

[0004] In the temperature sensor, a comparator outputs a first readout number when the voltage at a first input terminal is lower than the voltage at the second input terminal, and the comparator outputs a second readout number when the voltage at the second input terminal is lower than the voltage at the first input terminal. When the comparator outputs the first readout number, a read circuit is in an A-phase state. When the comparator outputs the second readout number, the read circuit is in a B-phase state.

[0005] When the readout circuit is in the A-phase state, a proportional current IPTAT is transmitted to a first end of a capacitor. Further, a second end of the capacitor is connected to a ground. Further, a voltage, which is the potential difference from the first to second ends of the capacitor, is input to the first input terminal of the comparator. Further, the voltage between the base and emitter of a bipolar junction transistor is input to the second input terminal of the comparator.

[0006] When the readout circuit is in the B-phase state, the proportional current IPTAT is transmitted to the second end of the capacitor. Further, the first end of the capacitor is connected to the ground. Further, ground voltage is input to the first input terminal of the comparator. Further, a voltage, which is the potential difference from the second to first ends of the capacitor, is input to the second input terminal of the comparator.

[0007] A controller determines a temperature based on the oscillation period of the readout number output by the comparator (paragraphs [0036], [0046], [0047], [0052], [0057], [0058], [0059], [0062], and [0063], as well as FIGS. 4 and 5).

SUMMARY OF THE INVENTION

[0008] In the temperature sensor disclosed in United States Unexamined Patent Application Publication No. 2022/0163402, the capacitor's second end receives a negative voltage when the readout circuit is in the B-phase state. This makes it difficult to form the temperature sensor onto a semiconductor substrate.

[0009] Further, in the temperature sensor, the proportional current IPTAT, which is transmitted to the capacitor's second end when the readout circuit is in the B-phase state, depends on the temperature property of a resistor provided in the circuit that generates the proportional current IPTAT. This makes it impossible to determine the temperature with high accuracy.

[0010] One aspect of the present disclosure has been made in view of this problem. The aspect of the present disclosure aims to provide a temperature measuring circuit that can be easily formed on a semiconductor substrate, and that can measure a temperature with high accuracy.

[0011] A temperature measuring circuit according to one aspect of the present disclosure is provided with the following: a capacitor configured to generate an inter-terminal voltage corresponding to the amount of stored electric charge; a first current supply configured to feed a first current; a second current supply configured to feed a second current; a first voltage supply configured to generate a first reference voltage; a second voltage supply configured to generate a second reference voltage; a third voltage supply configured to generate a third reference voltage corresponding to a temperature; and a comparator circuit configured to output an output voltage indicating the levels of two input voltages that are received. The temperature measuring circuit is also provided with a first switch circuit configured to, upon receiving a first control signal, charge the capacitor through the first current and set the two input voltages at a selected reference voltage and the inter-terminal voltage, and upon receiving a second control signal, discharge the capacitor through the second current and set the two input voltages at the second reference voltage and the inter-terminal voltage. The temperature measuring circuit is also provided with a second switch circuit configured to switch between a first connection state in which the selected reference voltage is set at the first reference voltage, and a second connection state in which the selected reference voltage is set at the third reference voltage. The temperature measuring circuit is also provided with a control circuit configured to, during a first period, set a control signal at the second control signal in response to the output voltage indicating that the inter-terminal voltage is higher than the first reference voltage while the control signal is set at the first control signal and the second switch circuit is in the first connection state, and set the control signal at the first control signal and bring the second switch circuit into the first connection state in response to the output voltage indicating that the inter-terminal voltage is lower than the second reference voltage while the control signal is set at the second control signal, and during a second period, set the control signal at the second control signal in response to the output voltage indicating that the inter-terminal voltage is higher than the third reference voltage while the control signal is set at the first control signal and the second switch circuit is in the second connection state, and set the control signal at the first control signal and bring the second switch circuit into the second connection state in response to the output voltage indicating that the inter-terminal voltage is lower than the second reference voltage while the control signal is set at the second control signal. The temperature measuring circuit is also provided with the following: a counter circuit configured to count a first frequency of the control signal during the first period, and a second frequency of the control signal during the second period; and a temperature obtaining unit configured to obtain the temperature based on the first and second frequencies.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1A is a circuit diagram illustrating a temperature measuring circuit in State 1 according to a first embodiment;

[0013] FIG. 1B is a circuit diagram illustrating the temperature measuring circuit in State 2 according to the first embodiment;

[0014] FIG. 1C is a circuit diagram illustrating the temperature measuring circuit in State 3 according to the first embodiment;

[0015] FIG. 1D is a circuit diagram illustrating the temperature measuring circuit in State 4 according to the first embodiment;

[0016] FIG. 2A is a flowchart showing the operation of the temperature measuring circuit according to the first embodiment;

[0017] FIG. 2B is a flowchart showing the operation of the temperature measuring circuit according to the first embodiment;

[0018] FIG. 3A is a timing chart showing the following time variations during a first period T1: time variations of a first input voltage Vin1 and a second input voltage Vin2, which are input to a comparator provided in the temperature measuring circuit according to the first embodiment; time variations of an inter-terminal voltage Vc, which is generated by a capacitor provided in the temperature measuring circuit; and time variations of a control signal S1, which is output by a control circuit provided in the temperature measuring circuit;

[0019] FIG. 3B is a timing chart showing the following time variations during a second period T2: time variations of the first input voltage Vin1 and the second input voltage Vin2, which are input to the comparator provided in the temperature measuring circuit according to the first embodiment; time variations of the inter-terminal voltage Vc, which is generated by the capacitor provided in the temperature measuring circuit; and time variations of the control signal S1, which is output by the control circuit provided in the temperature measuring circuit;

[0020] FIG. 4 is a circuit diagram of a third voltage supply provided in the temperature measuring circuit according to the first embodiment;

[0021] FIG. 5A is a circuit diagram illustrating a temperature measuring circuit in State 1a according to a second embodiment;

[0022] FIG. 5B is a circuit diagram illustrating the temperature measuring circuit in State 2a according to the second embodiment;

[0023] FIG. 5C is a circuit diagram illustrating the temperature measuring circuit in State 1b according to the second embodiment;

[0024] FIG. 5D is a circuit diagram illustrating the temperature measuring circuit in State 2b according to the second embodiment;

[0025] FIG. 5E is a circuit diagram illustrating the temperature measuring circuit in State 3a according to the second embodiment;

[0026] FIG. 5F is a circuit diagram illustrating the temperature measuring circuit in State 4a according to the second embodiment;

[0027] FIG. 5G is a circuit diagram illustrating the temperature measuring circuit in State 3b according to the second embodiment;

[0028] FIG. 5H is a circuit diagram illustrating the temperature measuring circuit in State 4b according to the second embodiment;

[0029] FIG. 6A is a circuit diagram illustrating a comparator circuit, which is provided in the temperature measuring

circuit according to the second embodiment, with its first and second input terminals being a non-inverting input terminal and an inverting input terminal, respectively;

[0030] FIG. 6B is a circuit diagram illustrating the comparator circuit, which is provided in the temperature measuring circuit according to the second embodiment, with its first and second input terminals being an inverting input terminal and a non-inverting input terminal, respectively;

[0031] FIG. 7A is a flowchart showing the operation of the temperature measuring circuit according to the second embodiment;

[0032] FIG. 7B is a flowchart showing the operation of the temperature measuring circuit according to the second embodiment;

[0033] FIG. 8A is a timing chart showing the following time variations during the first period T1: time variations of the first input voltage Vin1 and the second input voltage Vin2, which are input to a comparator provided in the temperature measuring circuit according to the second embodiment; time variations of the inter-terminal voltage Vc, which is generated by a capacitor provided in the temperature measuring circuit; time variations of the control signal S1, which is output by a control circuit provided in the temperature measuring circuit; and time variations of a control signal S3, which is output by a switching control circuit provided in the temperature measuring circuit; and

[0034] FIG. 8B is a timing chart showing the following time variations during the second period T2: time variations of the first input voltage Vin1 and the second input voltage Vin2, which are input to the comparator provided in the temperature measuring circuit according to the second embodiment; time variations of the inter-terminal voltage Vc, which is generated by the capacitor provided in the temperature measuring circuit; time variations of the control signal S1, which is output by the control circuit provided in the temperature measuring circuit; and time variations of the control signal S3, which is output by the switching control circuit provided in the temperature measuring circuit.

DETAILED DESCRIPTION OF THE INVENTION

[0035] The embodiments of the present disclosure will be described with reference to the drawings. It is noted that identical or equivalent constituents will be denoted by the same signs throughout the drawings, and the descriptions of redundancies will be omitted.

1 First Embodiment

1.1 Temperature Measuring Circuit

[0036] FIG. 1A is a circuit diagram illustrating a temperature measuring circuit in State 1 according to a first embodiment. FIG. 1B is a circuit diagram illustrating the temperature measuring circuit in State 2 according to the first embodiment. FIG. 1C is a circuit diagram illustrating the temperature measuring circuit in State 3 according to the first embodiment. FIG. 1D is a circuit diagram illustrating the temperature measuring circuit in State 4 according to the first embodiment.

[0037] The temperature measuring circuit, 1, according to the first embodiment illustrated in FIGS. 1A to 1D measures a temperature and outputs the measured temperature.

[0038] The temperature measuring circuit 1 is formed on a semiconductor substrate and is incorporated in a semiconductor integrated circuit. The semiconductor substrate is a complementary metal oxide semiconductor (CMOS) wafer for instance. The temperature measuring circuit 1 does not need to be formed on a semiconductor substrate and does not need to be incorporated in a semiconductor integrated circuit.

[0039] As illustrated in FIGS. 1A to 1D, the temperature measuring circuit 1 includes a capacitor 11, a first current supply 21, a second current supply 22, a first voltage supply 31, a second voltage supply 32, a third voltage supply 33, a comparator circuit 41, a control circuit 42, a counter circuit 43, a memory 44, a temperature obtaining unit 45, a first switch circuit 51, and a second switch circuit 52.

[0040] The capacitor 11 includes terminals 11a and 11b. The first current supply 21 includes a negative electrode 21a and a positive electrode 21b. The second current supply 22 includes a negative electrode 22a and a positive electrode 22b. The first voltage supply 31 includes a negative electrode 31a and a positive electrode 31b. The second voltage supply 32 includes a negative electrode 32a and a positive electrode 32b. The third voltage supply 33 includes a negative electrode 33a and a positive electrode 33b. The comparator circuit 41 includes a first input terminal 41a, a second input terminal 41b, and an output terminal 41c. The control circuit 42 includes an input terminal 42a and an output terminal 42b. The first switch circuit 51 includes terminals 51a, 51b, 51c, 51d, 51e, 51f, 51g, 51h, and 51i. The second switch circuit 52 includes terminals 52a, 52b, and 52c.

[0041] The terminal 11b of the capacitor 11 is grounded. The terminal 11a of the capacitor 11 is electrically connected to the terminal 51c of the first switch circuit 51.

[0042] The negative electrode 21a of the first current supply 21 is electrically connected to a power supply. The positive electrode 21b of the first current supply 21 is electrically connected to the terminal 51a of the first switch circuit 51. The negative electrode 22a of the second current supply 22 is electrically connected to the terminal 51b of the first switch circuit 51. The positive electrode 22b of the second current supply 22 is grounded. The negative electrode 31a of the first voltage supply 31 is grounded. The positive electrode 31b of the first voltage supply 31 is electrically connected to the terminal 52a of the second switch circuit 52. The negative electrode 32a of the second voltage supply 32 is grounded. The positive electrode 32b of the second voltage supply 32 is electrically connected to the terminal 51h of the first switch circuit 51. The negative electrode 33a of the third voltage supply 33 is grounded. The positive electrode 33b of the third voltage supply 33 is electrically connected to the terminal 52b of the second switch circuit 52. The first input terminal 41a and second input terminal 41b of the comparator circuit 41 are electrically connected to the terminals 51f and 51i of the first switch circuit 51, respectively. The output terminal 41c of the comparator circuit 41 is electrically connected to the input terminal 42a of the control circuit 42. The output terminal 42b of the control circuit 42 is electrically connected to the counter circuit 43, the first switch circuit 51, and the second switch circuit 52. The terminal 52c of the second switch circuit 52 is electrically connected to the terminal 51d of the first switch circuit 51. The counter circuit 43 is electrically connected to the control circuit 42. The

memory 44 is electrically connected to the counter circuit 43. The temperature obtaining unit 45 is electrically connected to the counter circuit 43.

[0043] The capacitor 11 stores electric charge. The amount of stored electric charge increases by the amount of electric charge carried by a current flowing into the terminal 11a of the capacitor 11 and decreases by the amount of electric charge carried by a current flowing out of the terminal 11a of the capacitor 11. The capacitor 11 generates an inter-terminal voltage V_c corresponding to the amount of stored electric charge between the terminal 11b and the terminal 11a.

[0044] The first current supply 21 feeds a first current I_1 flowing into the negative electrode 21a and flowing out of the positive electrode 21b.

[0045] The second current supply 22 feeds a second current I_2 flowing into the negative electrode 22a and flowing out of the positive electrode 22b.

[0046] The first current supply 21 and the second current supply 22 are temperature-proportional current supplies, temperature-insensitive current supplies, process-variations-insensitive current supply, or other things. The first current supply 21 and the second current supply 22 that are process-variations-insensitive current supplies enable a temperature to be measured with high accuracy even when there are process variations in the semiconductor substrate on which the temperature measuring circuit 1 is formed.

[0047] The first voltage supply 31 generates a first reference voltage V_{ref} between the negative electrode 31a and the positive electrode 31b.

[0048] The second voltage supply 32 generates a second reference voltage $V_{ref/M}$ between the negative electrode 32a and the positive electrode 32b.

[0049] The third voltage supply 33 generates a third reference voltage V_{be} between the negative electrode 33a and the positive electrode 33b.

[0050] The first reference voltage V_{ref} and the second reference voltage $V_{ref/M}$ are stable with respect to temperature. The third reference voltage V_{be} is a voltage corresponding to temperature. The second reference voltage $V_{ref/M}$ is lower than the first reference voltage V_{ref} and is $1/M$ times of the first reference voltage V_{ref} . M is a positive value greater than 1.

[0051] The comparator circuit 41 outputs, from the output terminal 41c, an output voltage V_{out} indicating the levels of a first input voltage V_{in1} input to the first input terminal 41a, and a second input voltage V_{in2} input to the second input terminal 41b. The first input terminal 41a and second input terminal 41b of the comparator circuit 41 are a non-inverting input terminal and an inverting input terminal, respectively. The comparator circuit 41 thus outputs a first output voltage High as the output voltage V_{out} when the first input voltage V_{in1} is higher than the second input voltage V_{in2} , and the comparator circuit 41 outputs a second output voltage Low as the output voltage V_{out} when the first input voltage V_{in1} is lower than the second input voltage V_{in2} .

[0052] The control circuit 42 receives the output voltage V_{out} , which is output from the output terminal 41c of the comparator circuit 41, from the input terminal 42a. The control circuit 42 outputs the control signals S1 and S2 from the output terminal 42b. The control circuit 42 switches the outgoing control signal S1 between a first control signal High and a second control signal Low in synchronization with changes of the received output voltage V_{out} from the

first output voltage High to the second output voltage Low. The control circuit 42 switches the outgoing control signal S2 between the first control signal High and the second control signal Low.

[0053] The counter circuit 43 receives the control signal S1 output from the output terminal 42b of the control circuit 42. The counter circuit 43 counts the frequency of the received control signal S1 and outputs the counted frequency.

[0054] The temperature obtaining unit 45 receives the frequency output from the counter circuit 43. The temperature obtaining unit 45 obtains a temperature from the received frequency.

[0055] The first switch circuit 51 receives the control signal S1 output from the output terminal 42b of the control circuit 42.

[0056] Upon receiving the control signal S1 being the first control signal High, the first switch circuit 51 brings its terminal 51a into electrical continuity with its terminal 51c and does not bring its terminal 51b into electrical continuity with its terminal 51c, as illustrated in FIG. 1A and FIG. 1C. The first switch circuit 51 thus causes the first current I1 flowing out of the positive electrode 21b of the first current supply 21 to flow into the terminal 11a of the capacitor 11, thus charging the capacitor 11 through the first current I1. This raises the inter-terminal voltage Vc, which is generated by the capacitor 11, along with time.

[0057] Upon receiving the control signal S1 being the second control signal Low, the first switch circuit 51 brings its terminal 51b into electrical continuity with its terminal 51c and does not bring its terminal 51a into electrical continuity with its terminal 51c, as illustrated in FIGS. 1B and 1D. The first switch circuit 51 thus causes the second current I2 flowing into the negative electrode 22a of the second current supply 22 to flow out of the terminal 11a of the capacitor 11, thus discharging the capacitor 11 through the second current I2. This lowers the inter-terminal voltage Vc, which is generated by the capacitor 11, along with time.

[0058] In addition, upon receiving the control signal S1 being the first control signal High, the first switch circuit 51 brings its terminal 51d into electrical continuity with its terminal 51f and does not bring its terminal 51e into electrical continuity with its terminal 51f, as illustrated in FIGS. 1A and 1C. The first switch circuit 51 thus sets the first input voltage Vin1 input to the first input terminal 41a of the comparator circuit 41, at the reference voltage Vref or Vbe selected by the second switch circuit 52.

[0059] Upon receiving the control signal S1 being the second control signal Low, the first switch circuit 51 brings its terminal 51e into electrical continuity with its terminal 51f and does not bring its terminal 51d into electrical continuity with its terminal 51f, as illustrated in FIGS. 1B and 1D. The first switch circuit 51 thus sets the first input voltage Vin1 at the inter-terminal voltage Vc generated by the capacitor 11.

[0060] In addition, upon receiving the control signal S1 being the first control signal High, the first switch circuit 51 brings its terminal 51g into electrical continuity with its terminal 51i and does not bring its terminal 51h into electrical continuity with its terminal 51i, as illustrated in FIGS. 1A and 1C. The first switch circuit 51 thus sets the second input voltage Vin2 at the inter-terminal voltage Vc generated by the capacitor 11.

[0061] Upon receiving the control signal S1 being the second control signal Low, the first switch circuit 51 brings its terminal 51h into electrical continuity with its terminal 51i and does not bring its terminal 51g into electrical continuity with its terminal 51i, as illustrated in FIGS. 1B and 1D. The first switch circuit 51 thus sets the second input voltage Vin2 at the second reference voltage Vref/M generated by the second voltage supply 32.

[0062] As described above, upon receiving the control signal S1 being the first control signal High, the first switch circuit 51 sets the two input voltages Vin1 and Vin2, which are compared by the comparator circuit 41, at the reference voltage Vref or Vbe, which is selected by the second switch circuit 52, and the inter-terminal voltage Vc, which is generated by the comparator circuit 11. Upon receiving the control signal S1 being the second control signal Low, the first switch circuit 51 sets the two input voltages Vin1 and Vin2, which are compared by the comparator circuit 41, at the inter-terminal voltage Vc, which is generated by the capacitor 11, and the second reference voltage Vref/M, which is generated by the second voltage supply 32.

[0063] The second switch circuit 52 receives the control signal S2 output from the output terminal 42b of the control circuit 42.

[0064] Upon receiving the control signal S2 being the first control signal High, the second switch circuit 52 brings its terminal 52a into electrical continuity with its terminal 52c and does not bring its terminal 52b into electrical continuity with its terminal 52c, as illustrated in FIG. 1A, FIG. 1B, and FIG. 1D. The second switch circuit 52 thus selects the first reference voltage Vref generated by the first voltage supply 31.

[0065] Upon receiving the control signal S2 being the second control signal Low, the second switch circuit 52 brings its terminal 52b into electrical continuity with its terminal 52c and does not bring its terminal 52a into electrical continuity with its terminal 52c, as illustrated in FIG. 1C. The second switch circuit 52 thus selects the third reference voltage Vref generated by the third voltage supply 33.

[0066] As described above, the second switch circuit 52 switches between a first connection state in which a reference voltage selected by the second switch circuit 52 is the first reference voltage Vref generated by the first voltage supply 31, and a second connection state in which the reference voltage selected by the second switch circuit 52 is the third reference voltage Vbe generated by the third voltage supply 33.

1.2 Operation of Temperature Measuring Circuit

[0067] FIGS. 2A and 2B are flowcharts showing the operation of the temperature measuring circuit according to the first embodiment.

[0068] The temperature measuring circuit 1 executes Steps S101 to S114 shown in FIGS. 2A and 2B.

[0069] The temperature measuring circuit 1 executes Steps S101 to S106 during a first period T1, which has a first measurement time TF1, to turn alternately into State 1 illustrated in FIG. 1A and State 2 illustrated in FIG. 1B. The temperature measuring circuit 1 executes Steps S108 to S113 during a subsequent second period T2, which has a second measurement time TF2, to turn alternately into State 3 illustrated in FIG. 1C and State 4 illustrated in FIG. 1D.

[0070] The temperature measuring circuit 1 executes Steps S101 to S103 during a period T1A, which is included in the first period T1, to turn into State 1 illustrated in FIG. 1A. The temperature measuring circuit 1 executes Steps S104 to S106 during a period T1B, which is included in the first period T1, to turn into State 2 illustrated in FIG. 1B.

[0071] The temperature measuring circuit 1 executes Steps S108 to S110 during a period T2A, which is included in the second period T2, to turn into State 3 illustrated in FIG. 1C. The temperature measuring circuit 1 executes Steps S111 to S113 during a period T2B, which is included in the second period T2, to turn into State 4 illustrated in FIG. 1D.

Charge during Period T1A

[0072] In Step S101, as illustrated in FIG. 1A, the control circuit 42 sets the control signal S1 at the first control signal High and sets the control signal S2 at the first control signal High.

[0073] Accordingly, the positive electrode 21b of the first current supply 21 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the positive electrode 31b of the first voltage supply 31. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11.

[0074] Accordingly, the first current supply 21 charges the capacitor 11. Further, the comparator circuit 41 compares the first reference voltage Vref and the inter-terminal voltage Vc and sets the output voltage Vout at the first output voltage High.

[0075] In Step S102, the control circuit 42 determines whether a time TM1 elapsed from the start of the first period T1 is shorter than the first measurement time TF1. If determining that the time TM1 is shorter than the first measurement time TF1, the control circuit 42 executes Step S103. If determining that the time TM1 is equal to or longer than the first measurement time TF1, the control circuit 42 executes Step S107.

[0076] In Step S103, the control circuit 42 determines whether the inter-terminal voltage Vc is lower than the first reference voltage Vref on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is lower than the first reference voltage Vref, the control circuit 42 executes Step S101. If determining that the inter-terminal voltage Vc is equal to or higher than the first reference voltage Vref, the control circuit 42 executes Step S104.

[0077] From Steps S101 to S103, the control circuit 42 causes the first current supply 21 to continue charging the capacitor 11 and causes the comparator circuit 41 to continue comparing the first reference voltage Vref and the inter-terminal voltage Vc until the time TM1 reaches the first measurement time TF1, or until the inter-terminal voltage Vc rises to the first reference voltage Vref. The control circuit 42 ends the first period T1 and starts the second period T2 in synchronization with that the time TM1 reaches the first measurement time TF1. The control circuit 42 ends the period T1A and starts the period T1B in synchronization with that the inter-terminal voltage Vc rises to the first reference voltage Vref, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Discharge during Period T1B

[0078] In Step S104, as illustrated in FIG. 1B, the control circuit 42 sets the control signal S1 at the second control signal Low and sets the control signal S2 at the first control signal High. The control circuit 42 may set the control signal S2 at the second control signal Low.

[0079] Accordingly, the negative electrode 22a of the second current supply 22 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the positive electrode 32b of the second voltage supply 32.

[0080] Accordingly, the second current supply 22 discharges the capacitor 11. Further, the comparator circuit 41 compares the inter-terminal voltage Vc and the second reference voltage Vref/M and sets the output voltage Vout at the first output voltage High.

[0081] In subsequent Step S105, the control circuit 42 determines whether the time TM1 is shorter than the first measurement time TF1. If determining that the time TM1 is shorter than the first measurement time TF1, the control circuit 42 executes Step S106. If determining that the time TM1 is equal to or longer than the first measurement time TF1, the control circuit 42 executes Step S107.

[0082] In Step S106, the control circuit 42 determines whether the inter-terminal voltage Vc is higher than the second reference voltage Vref/M on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is higher than the second reference voltage Vref/M, the control circuit 42 executes Step S104. If determining that the inter-terminal voltage Vc is equal to or lower than the second reference voltage Vref/M, the control circuit 42 executes Step S101.

[0083] From Steps S104 to S106, the control circuit 42 causes the second current supply 22 to continue discharging the capacitor 11 and causes the comparator circuit 41 to continue comparing the inter-terminal voltage Vc and the second reference voltage Vref/M until the time TM1 reaches the first measurement time TF1, or until the inter-terminal voltage Vc lowers to the second reference voltage Vref/M. The control circuit 42 ends the first period T1 and starts the second period T2 in synchronization with that the time TM1 reaches the first measurement time TF1. The control circuit 42 ends the period T1B and starts the period T1A in synchronization with that the inter-terminal voltage Vc lowers to the second reference voltage Vref/M, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Counting During First Period T1

[0084] From Steps S101 to S106, the control circuit 42 sets the control signal S1 at the second control signal Low in response to the output voltage Vout indicating that the inter-terminal voltage Vc has risen to the first reference voltage Vref while the control signal S1 is the first reference voltage Vref, and while the second switch circuit 52 is in the first connection state. The control circuit 42 sets the control signal S1 at the first control signal High and brings the second switch circuit 52 into the first connection state in response to the output voltage Vout indicating that the inter-terminal voltage Vc has lowered to the second reference voltage Vref/M while the control signal S1 is the second control signal Low.

[0085] That the inter-terminal voltage V_c has risen to the first reference voltage V_{ref} is indicated by a change of the output voltage V_{out} from the first output voltage High to the second output voltage Low. That the inter-terminal voltage V_c has lowered to the second output voltage Low is indicated by a change of the output voltage V_{out} from the first output voltage High to the second output voltage Low.

[0086] As such, the control signal $S1$ changes between the first control signal High and the second control signal Low periodically during the first period $T1$. The counter circuit 43 counts how many times the control signal $S1$ has changed between the first control signal High and the second control signal Low during the first period $T1$ (hereinafter, referred to as “counts”), and the counter circuit 43 writes the counts into a first address of the memory 44.

Reset of Counter Circuit and Change of Count Writing Destination

[0087] In Step S107, the counter circuit 43 is reset, and the destination into which the foregoing number of periodic changes counted by the counter circuit 43 is written is changed from the first address to a second address.

Charge During Period T2A

[0088] In Step S108, the control circuit 42 sets the control signal $S1$ at the first control signal High and sets the control signal $S2$ at the second control signal Low, as illustrated in FIG. 1C.

[0089] Accordingly, the positive electrode 21b of the first current supply 21 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the positive electrode 33b of the third voltage supply 33. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11.

[0090] Accordingly, the first current supply 21 charges the capacitor 11. Further, the comparator circuit 41 compares the third reference voltage V_{be} and the inter-terminal voltage V_c and sets the output voltage V_{out} at the first output voltage High.

[0091] In subsequent Step S109, the control circuit 42 determines whether a time $TM2$ elapsed from the start of the second period $T2$ is shorter than the second measurement time $TF2$. If determining that the time $TM2$ is shorter than the second measurement time $TF2$, the control circuit 42 executes Step S110. If determining that the time $TM2$ is equal to or longer than the second measurement time $TF2$, the control circuit 42 executes Step S114.

[0092] In Step S110, the control circuit 42 determines whether the inter-terminal voltage V_c is lower than the third reference voltage V_{be} on the basis of the output voltage V_{out} . If determining that the inter-terminal voltage V_c is lower than the third reference voltage V_{be} , the control circuit 42 executes Step S108. If determining that the inter-terminal voltage V_c is equal to or higher than the third reference voltage V_{be} , the control circuit 42 executes Step S111.

[0093] From Steps S108 to S110, the control circuit 42 causes the first current supply 21 to continue charging the capacitor 11 and causes the comparator circuit 41 to continue comparing the third reference voltage V_{be} and the inter-terminal voltage V_c until the time $TM2$ reaches the

second measurement time $TF2$, or until the inter-terminal voltage V_c rises to the third reference voltage V_{be} . The control circuit 42 ends the second period $T2$ in synchronization with that the time $TM2$ reaches the second measurement time $TF2$. The control circuit 42 ends the period $T2A$ and starts the period $T2B$ in synchronization with that the inter-terminal voltage V_c rises to the third reference voltage V_{be} , and that the output voltage V_{out} changes from the first output voltage High to the second output voltage Low.

Discharge During Period T2B

[0094] In Step S111, as illustrated in FIG. 1D, the control circuit 42 sets the control signal $S1$ at the second control signal Low and sets the control signal $S2$ at the first control signal High. The control circuit 42 may set the control signal $S2$ at the second control signal Low.

[0095] Accordingly, the negative electrode 22a of the second current supply 22 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the positive electrode 32b of the second voltage supply 32.

[0096] Accordingly, the second current supply 22 discharges the capacitor 11. Further, the comparator circuit 41 compares the inter-terminal voltage V_c and the second reference voltage V_{ref}/M and sets the output voltage V_{out} at the first output voltage High.

[0097] In subsequent Step S112, the control circuit 42 determines whether the time $TM2$ is shorter than the second measurement time $TF2$. If determining that the time $TM2$ is shorter than the second measurement time $TF2$, the control circuit 42 executes Step S113. If determining that the time $TM2$ is equal to or longer than the second measurement time $TF2$, the control circuit 42 executes Step S114.

[0098] In Step S113, the control circuit 42 determines whether the inter-terminal voltage V_c is higher than the second reference voltage V_{ref}/M on the basis of the output voltage V_{out} . If determining that the inter-terminal voltage V_c is higher than the second reference voltage V_{ref}/M , the control circuit 42 executes Step S111. If determining that the inter-terminal voltage V_c is equal to or lower than the second reference voltage V_{ref}/M , the control circuit 42 executes Step S108.

[0099] From Steps S111 to S113, the control circuit 42 causes the second current supply 22 to continue discharging the capacitor 11 and causes the comparator circuit 41 to continue comparing the inter-terminal voltage V_c and the second reference voltage V_{ref}/M until the time $TM2$ reaches the second measurement time $TF2$, or until the inter-terminal voltage V_c lowers to the second reference voltage V_{ref}/M . The control circuit 42 ends the second period $T2$ in synchronization with that the time $TM2$ reaches the second measurement time $TF2$. The control circuit 42 ends the period $T2B$ and starts the period $T2A$ in synchronization with that the inter-terminal voltage V_c lowers to the second reference voltage V_{ref}/M , and that the output voltage V_{out} changes from the first output voltage High to the second output voltage Low.

Counting During Second Period T2

[0100] From Steps S107 to S113, the control circuit 42 sets the control signal $S1$ at the second control signal Low in

response to the output voltage V_{out} indicating that the inter-terminal voltage V_c has risen to the third reference voltage V_{be} while the control signal $S1$ is the first control signal High, and while the second switch circuit **52** is in the first connection state. The control circuit **42** sets the control signal $S1$ at the first control signal High and brings the second switch circuit **52** into the first connection state in response to the output voltage V_{out} indicating that the inter-terminal voltage V_c has lowered to the second reference voltage V_{ref}/M while the control signal $S1$ is the second control signal Low.

[0101] That the inter-terminal voltage V_c has risen to the third reference voltage V_{be} is indicated by a change of the output voltage V_{out} from the first output voltage High to the second output voltage Low. That the inter-terminal voltage V_c has lowered to the second reference voltage V_{ref}/M is indicated by a change of the output voltage V_{out} from the first output voltage High to the second output voltage Low.

[0102] As such, the control signal $S1$ changes between the first control signal High and the second control signal Low periodically during the second period $T2$. The counter circuit **43** counts how many times the control signal $S1$ has changed between the first control signal High and the second control signal Low during the second period $T2$, and the counter circuit **43** writes the counts into the second address of the memory **44**.

Counts Readout

[0103] In Step $S114$, the counter circuit **43** reads the counts from the first and second addresses, and from the counts read from the first and second addresses, the counter circuit **43** obtains a first frequency $F1$ of the control signal $S1$ during the first period $T1$, and a second frequency $F2$ of the control signal $S1$ during the second period $T2$. Further, the temperature obtaining unit **45** obtains the temperature on the basis of the first frequency $F1$ and second frequency $F2$.

1.3 Time Variations of First Input Voltage, Second Input Voltage, Inter-Terminal Voltage, and Control Signal

Time Variations During First Period $T1$

[0104] FIG. 3A is a timing chart showing the following time variations during the first period $T1$: time variations of the first input voltage V_{in1} and second input voltage V_{in2} , which are input to the comparator provided in the temperature measuring circuit according to the first embodiment; time variations of the inter-terminal voltage V_c , which is generated by the capacitor provided in the temperature measuring circuit; and time variations of the control signal $S1$, which is output by the control circuit provided in the temperature measuring circuit.

[0105] As shown in FIG. 3A, in the temperature measuring circuit **1** in Steps $S101$ to $S106$, the periods $T1A$ and $T1B$ come alternately during the first period $T1$.

[0106] During the period $T1A$, the first input voltage V_{in1} and the second input voltage V_{in2} are at the first reference voltage V_{ref} and the inter-terminal voltage V_c , respectively. In addition, the inter-terminal voltage V_c rises along with time. Accordingly, the inter-terminal voltage V_c rises from the second reference voltage V_{ref}/M to the first reference voltage V_{ref} . In addition, the control signal $S1$ is kept at the first control signal High.

[0107] During the period $T1B$, the first input voltage V_{in1} and the second input voltage V_{in2} are at the inter-terminal voltage V_c and the second reference voltage V_{ref}/M , respectively. In addition, the inter-terminal voltage V_c lowers along with time. Accordingly, the inter-terminal voltage V_c lowers from the first reference voltage V_{ref} to the second reference voltage V_{ref}/M . In addition, the control signal $S1$ is kept at the second control signal Low.

[0108] During the periods $T1A$ and $T1B$, the input terminal to which the inter-terminal voltage V_c is input is switched between the first input terminal **41a** and second input terminal **41b** of the comparator circuit **41**. This enables switching between a state in which the fact that the inter-terminal voltage V_c has risen to a reference voltage can be detected, and a state in which the fact that the inter-terminal voltage V_c has lowered to the reference voltage can be detected. Also, during the periods $T1A$ and $T1B$, the reference voltage compared to the inter-terminal voltage V_c is switched between the first reference voltage V_{ref} and the second reference voltage V_{ref}/M . This enables switching between a state in which the fact that the inter-terminal voltage V_c has risen to the first reference voltage V_{ref} can be detected, and a state in which the fact that the inter-terminal voltage V_c has lowered to the second reference voltage V_{ref}/M can be detected.

[0109] The time necessary for the inter-terminal voltage V_c to rise from the second reference voltage V_{ref}/M to the first reference voltage V_{ref} does not depend on temperature. The time necessary for the inter-terminal voltage V_c to lower from the first reference voltage V_{ref} to the second reference voltage V_{ref}/M does not depend on temperature. Hence, the first frequency $F1$ is independent of temperature and can be used as a reference frequency.

Time Variations During Second Period $T2$

[0110] FIG. 3B is a timing chart showing the following time variations during the second period $T2$: time variations of the first input voltage V_{in1} and second input voltage V_{in2} , which are input to the comparator provided in the temperature measuring circuit according to the first embodiment; time variations of the inter-terminal voltage V_c , which is generated by the capacitor provided in the temperature measuring circuit; and time variations of the control signal $S1$, which is output by the control circuit provided in the temperature measuring circuit.

[0111] As shown in FIG. 3B, in the temperature measuring circuit **1** in Steps $S108$ to $S113$, the periods $T2A$ and $T2B$ come alternately during the second period $T2$.

[0112] During the period $T2A$, the first input voltage V_{in1} and the second input voltage V_{in2} are at the third reference voltage V_{be} and the inter-terminal voltage V_c , respectively. In addition, the inter-terminal voltage V_c rises along with time. Accordingly, the inter-terminal voltage V_c rises from the second reference voltage V_{ref}/M to the third reference voltage V_{be} . In addition, the control signal $S1$ is kept at the first control signal High.

[0113] During the period $T2B$, the first input voltage V_{in1} and the second input voltage V_{in2} are at the inter-terminal voltage V_c and the second reference voltage V_{ref}/M , respectively. In addition, the inter-terminal voltage V_c lowers along with time. Accordingly, the inter-terminal voltage V_c lowers from the third reference voltage V_{be} to the second reference voltage V_{ref}/M . In addition, the control signal $S1$ is kept at the second control signal Low.

[0114] During the periods T2A and T2B, the input terminal to which the inter-terminal voltage Vc is input is switched between the first input terminal 41a and second input terminal 41b of the comparator circuit 41. This enables switching between a state in which the fact that the inter-terminal voltage Vc has risen to a reference voltage can be detected, and a state in which the fact that the inter-terminal voltage Vc has lowered to the reference voltage can be detected. Also, during the periods T2A and T2B, the reference voltage compared to the inter-terminal voltage Vc is switched between the third reference voltage Vbe and the second reference voltage Vref/M. This enables switching between a state in which the fact that the inter-terminal voltage Vc has risen to the third reference voltage Vbe can be detected, and a state in which the fact that the inter-terminal voltage Vc has lowered to the second reference voltage Vref/M can be detected.

[0115] The time necessary for the inter-terminal voltage Vc to rise from the second reference voltage Vref/M to the third reference voltage Vbe depends on temperature. The time necessary for the inter-terminal voltage Vc to lower from the third reference voltage Vbe to the second reference voltage Vref/M depends on temperature. Hence, the second frequency F2 is dependent on temperature, and unlike the first frequency F1, the second frequency F2 can be used for specifying a temperature.

1.4 First Reference Voltage, Second Reference Voltage, and Third Reference Voltage

[0116] The first voltage supply 31 includes a bandgap reference circuit. The bandgap reference circuit generates temperature-independent voltage. The first voltage supply 31 extracts the generated voltage from the bandgap reference circuit and uses the extracted voltage as the first reference voltage Vref.

[0117] The second voltage supply 32 includes a bandgap reference circuit and a resistive voltage-dividing circuit. The bandgap reference circuit generates temperature-independent voltage. The resistive voltage-dividing circuit divides the generated voltage. The second voltage supply 32 extracts the divided voltage from the resistive voltage-dividing circuit and uses the extracted voltage as the second reference voltage Vref/M.

[0118] FIG. 4 is a circuit diagram of the third voltage supply provided in the temperature measuring circuit according to the first embodiment.

[0119] As illustrated in FIG. 4, the third voltage supply 33 includes a current supply 61 and a bipolar transistor 62. The current supply 61 has a negative electrode 61a and a positive electrode 61b. The bipolar transistor 62 has a base 62a, a collector 62b, and an emitter 62c.

[0120] The negative electrode 61a of the current supply 61 is electrically connected to a power supply. The positive electrode 61b of the current supply 61 is electrically connected to the emitter 62c of the bipolar transistor 62. The bipolar transistor 62 is in diode connection. Thus, the base 62a and collector 62b of the bipolar transistor 62 are electrically connected to each other. The base 62a and the collector 62b are grounded.

[0121] The current supply 61 feeds a current flowing into the negative electrode 61a and flowing out of the positive electrode 61b. The current to be fed may or may not be temperature-proportional current. However, a semiconductor integrated circuit that incorporates the temperature mea-

suring circuit 1 includes, in most cases, a current supply that feeds a temperature-proportional current. Hence, when the current to be fed is temperature-proportional current, the semiconductor integrated circuit can be downsized.

[0122] The bipolar transistor 62 feeds a current flowing into the emitter 62c and flowing out of the collector 62b.

[0123] The third voltage supply 33 extracts a voltage generated between the base 62a and emitter 62c from the emitter 62c and uses the extracted voltage as the third reference voltage Vbe.

[0124] The temperature coefficient of the voltage generated between the base 62a and emitter 62c is significantly smaller than the temperature coefficient of temperature-proportional current. Hence, even when the current to be fed is temperature-proportional current, the voltage extracted from the emitter 62c has a temperature coefficient close to the former temperature coefficient.

1.5 Temperature Obtainment

[0125] A ratio (Vbe-Vref/M):(Vref-Vref/M) coincides with the ratio between the first frequency F1 and second frequency F2, i.e., F1:F2, where (Vbe-Vref/M) denotes the difference between the third reference voltage Vbe and second reference voltage Vref/M, and where (Vref-Vref/M) denotes the difference between the first reference voltage Vref and second reference voltage Vref/M. Accordingly, the temperature obtaining unit 45 determines the third reference voltage Vbe from the first reference voltage Vref, the second reference voltage Vref/M, the first frequency F1, and the second frequency F2 in accordance with Mathematical Expression (11).

[Numeral 1]

$$Vbe = \frac{F1}{F2} \times \left(Vref - \frac{Vref}{M} \right) + \frac{Vref}{M} \quad (11)$$

[0126] The temperature obtaining unit 45 also obtains a temperature Temp from the temperature property of the determined third reference voltage Vbe. For example, the temperature obtaining unit 45 obtains the temperature Temp in accordance with Mathematical Expression (12) using a Boltzmann constant K, a saturation current Is, and a collector current Ic.

[Numeral 2]

$$Temp = \frac{Vbe}{K} \cdot \log \left[\frac{Is + Ic}{Is} \right] \quad (12)$$

1.6 Advantages of Temperature Measuring Circuit

[0127] In the temperature measuring circuit 1, the inter-terminal voltage Vc is kept at the second reference voltage Vref/M, which is a positive voltage, or higher. Thus, the voltage at the terminal 11a of the capacitor 11 is not a positive voltage. This facilitates forming the temperature measuring circuit 1 onto the semiconductor substrate.

[0128] In the temperature measuring circuit 1, the first current supply 21 and the second current supply 22 do not have to be temperature-proportional current supplies. This enables various current supplies to be used as the first

current supply **21** and the second current supply **22**. This also enables the temperature measuring circuit **1** to measure the temperature with high accuracy without being affected by variations in temperature-proportional current.

2 Second Embodiment

[0129] The following describes points in which the second embodiment is different from the first embodiment. With regard to what will not be described, a configuration similar to the configuration applied to the first embodiment will be applied to the second embodiment as well.

2.1 Temperature Measuring Circuit

[0130] FIG. 5A is a circuit diagram illustrating a temperature measuring circuit in State **1b** according to the second embodiment. FIG. 5B is a circuit diagram illustrating the temperature measuring circuit in State **2a** according to the second embodiment. FIG. 5C is a circuit diagram illustrating the temperature measuring circuit in State **1b** according to the second embodiment. FIG. 5D is a circuit diagram illustrating the temperature measuring circuit in State **2b** according to the second embodiment. FIG. 5E is a circuit diagram illustrating the temperature measuring circuit in State **3a** according to the second embodiment. FIG. 5F is a circuit diagram illustrating the temperature measuring circuit in State **4a** according to the second embodiment. FIG. 5G is a circuit diagram illustrating the temperature measuring circuit in State **3b** according to the second embodiment. FIG. 5H is a circuit diagram illustrating the temperature measuring circuit in State **4b** according to the second embodiment.

[0131] The temperature measuring circuit, **2**, according to the second embodiment illustrated in FIGS. 5A to 5H includes a switching control circuit **53**.

[0132] The switching control circuit **53** receives the control signal **S1** output from the output terminal **42b** of the control circuit **42**. The switching control circuit **53** divides the received control signal **S1** and outputs a control signal **S3**. The switching control circuit **53** outputs the first control signal High or the second control signal Low as the control signal **S3**. The switching control circuit **53** sets the frequency of a change in the outgoing control signal **S3** between the first control signal High and the second control signal Low, at $1/N$ of the frequency of a change in the received control signal **S1** between the first control signal High and the second control signal Low. Here, N is an integer equal to or greater than 2. The switching control circuit **53** changes the outgoing control signal **S3** from the first control signal High to the second control signal Low when the control signal **S1** received with the outgoing control signal **S3** being the first control signal High is set at the first control signal High by the setpoint N . The switching control circuit **53** changes the outgoing control signal **S3** from the second control signal Low to the first control signal High when the control signal **S1** received with the outgoing control signal **S3** being the second control signal Low is set at the first control signal High by the setpoint N . The switching control circuit **53** is formed from a flip-flop circuit or other circuits.

[0133] The comparator circuit **41** in the temperature measuring circuit **2** receives the control signal **S3** output from the switching control circuit **53**. The comparator circuit **41** sets its first input terminal **41a** and second input terminal **41b**

as a non-inverting input terminal and an inverting input terminal, respectively, while the received control signal **S3** is the second control signal Low. The comparator circuit **41** sets its first input terminal **41a** and second input terminal **41b** as an inverting input terminal and a non-inverting input terminal, respectively, while the received control signal **S3** is the first control signal High. Thus, while the received control signal **S3** is the second control signal Low, the comparator circuit **41** sets the output voltage **Vout** at the first output voltage High in response to the first input voltage **Vin1** higher than the second input voltage **Vin2**, and the comparator circuit **41** sets as the output voltage **Vout** at the second output voltage Low in response to the first input voltage **Vin1** lower than the second input voltage **Vin2**. Further, while the received control signal **S3** is the first control signal High, the comparator circuit **41** sets the output voltage **Vout** at the second output voltage Low in response to the first input voltage **Vin1** higher than the second input voltage **Vin2**, and the comparator circuit **41** sets the output voltage **Vout** at the first output voltage High in response to the first input voltage **Vin1** lower than the second input voltage **Vin2**.

[0134] The first switch circuit **51** in the temperature measuring circuit **2** includes terminals **51j** and **51k**, as illustrated in FIGS. 5A through 5H.

[0135] The terminal **51j** of the first switch circuit **51** is electrically connected to the positive electrode **32b** of the second voltage supply **32**. The terminal **51k** of the first switch circuit **51** is electrically connected to the terminal **52c** of the second switch circuit **52**.

[0136] The first switch circuit **51** receives the control signal **S3** output from the switching control circuit **53**.

[0137] Upon receiving the control signal **S1** being the first control signal High, and receiving the control signal **S3** being the second control signal Low, the first switch circuit **51** brings its terminal **51d** into electrical continuity with its terminal **51f** and does not bring its terminals **51e** and **51j** into electrical continuity with its terminal **51f**, as illustrated in FIGS. 5A and 5E. The first switch circuit **51** thus sets the first input voltage **Vin1** at the reference voltage **Vref** or **Vbe** selected by the second switch circuit **52**.

[0138] Upon receiving the control signal **S1** being the second control signal Low, and receiving the control signal **S3** being the second control signal Low, the first switch circuit **51** brings its terminal **51e** into electrical continuity with its terminal **51f** and does not bring its terminals **51d** and **51j** into electrical continuity with its terminal **51f**, as illustrated in FIGS. 5B and 5F. The first switch circuit **51** thus sets the first input voltage **Vin1** at the inter-terminal voltage **Vc** generated by the capacitor **11**.

[0139] Upon receiving the control signal **S1** being the first control signal High, and receiving the control signal **S3** being the first control signal High, the first switch circuit **51** brings its terminal **51e** into electrical continuity with its terminal **51f** and does not bring its terminals **51d** and **51j** into electrical continuity with its terminal **51f**, as illustrated in FIGS. 5C and 5G. The first switch circuit **51** thus sets the first input voltage **Vin1** at the inter-terminal voltage **Vc** generated by the capacitor **11**.

[0140] Upon receiving the control signal **S1** being the second control signal Low, and receiving the control signal **S3** being the first control signal High, the first switch circuit **51** brings its terminal **51j** into electrical continuity with its terminal **51f** and does not bring its terminals **51d** and **51e**

into electrical continuity with its terminal **51f**, as illustrated in FIGS. **5D** and **5H**. The first switch circuit **51** thus sets the first input voltage **Vin1** at the second reference voltage **Vref/M** generated by the second voltage supply **32**.

[0141] Upon receiving the control signal **S1** being the first control signal High, and receiving the control signal **S3** being the second control signal Low, the first switch circuit **51** brings its terminal **51g** into electrical continuity with its terminal **51i** and does not bring its terminals **51h** and **51k** into electrical continuity with its terminal **51i**, as illustrated in FIGS. **5A** and **5E**. The first switch circuit **51** thus sets the second input voltage **Vin2** at the inter-terminal voltage **Vc** generated by the capacitor **11**.

[0142] Upon receiving the control signal **S1** being the second control signal Low, and receiving the control signal **S3** being the second control signal Low, the first switch circuit **51** brings its terminal **51h** into electrical continuity with its terminal **51i** and does not bring its terminals **51g** and **51k** into electrical continuity with its terminal **51i**, as illustrated in FIGS. **5B** and **5F**. The first switch circuit **51** thus sets the second input voltage **Vin2** at the second reference voltage **Vref/M** generated by the second voltage supply **32**.

[0143] Upon receiving the control signal **S1** being the first control signal High, and receiving the control signal **S3** being the first control signal High, the first switch circuit **51** brings its terminal **51k** into electrical continuity with its terminal **51i** and does not bring its terminals **51g** and **51h** into electrical continuity with its terminal **51i**, as illustrated in FIGS. **5C** and **5G**. The first switch circuit **51** thus sets the second input voltage **Vin2** at the reference voltage **Vref** or **Vbe** selected by the second switch circuit **52**.

[0144] Upon receiving the control signal **S1** being the second control signal Low, and receiving the control signal **S3** being the first control signal High, the first switch circuit **51** brings its terminal **51g** into electrical continuity with its terminal **51i** and does not bring its terminals **51h** and **51k** into electrical continuity with its terminal **51i**, as illustrated in FIGS. **5D** and **5H**. The first switch circuit **51** thus sets the second input voltage **Vin2** at the inter-terminal voltage **Vc** generated by the capacitor **11**.

[0145] As described above, upon receiving the control signal **S1** being the first control signal High, and receiving the control signal **S3** being the second control signal Low, the first switch circuit **51** sets the two input voltages **Vin1** and **Vin2**, which are compared by the comparator circuit **41**, at the reference voltage **Vref** or **Vbe**, which is selected by the second switch circuit **52**, and the inter-terminal voltage **Vc**, which is generated by the comparator circuit **11**. Upon receiving the control signal **S1** being the second control signal Low, and receiving the control signal **S3** being the second control signal Low, the first switch circuit **51** sets the two input voltages **Vin1** and **Vin2**, which are compared by the comparator circuit **41**, at the inter-terminal voltage **Vc**, which is generated by the capacitor **11**, and the second reference voltage **Vref/M**, which is generated by the second voltage supply **32**. Upon receiving the control signal **S1** being the first control signal High, and receiving the control signal **S3** being the first control signal High, the first switch circuit **51** sets the two input voltages **Vin1** and **Vin2**, which are compared by the comparator circuit **41**, at the inter-terminal voltage **Vc**, which is generated by the comparator circuit **11**, and the reference voltage **Vref** or **Vbe**, which is selected by the second switch circuit **52**. Upon receiving the control signal **S1** being the second control signal Low, and

receiving the control signal **S3** being the first control signal High, the first switch circuit **51** sets the two input voltages **Vin1** and **Vin2**, which are compared by the comparator circuit **41**, at the second reference voltage **Vref/M**, which is generated by the capacitor **32**, and the inter-terminal voltage **Vc**, which is generated by the second voltage supply **11**.

2.2 Comparator Circuit

[0146] FIG. **6A** is a circuit diagram illustrating the comparator circuit, which is provided in the temperature measuring circuit according to the second embodiment, with its first and second input terminals being a non-inverting input terminal and an inverting input terminal, respectively. FIG. **6B** is a circuit diagram illustrating the comparator circuit, which is provided in the temperature measuring circuit according to the second embodiment, with its first and second input terminals being an inverting input terminal and a non-inverting input terminal, respectively.

[0147] As illustrated in FIGS. **6A** and **6B**, the comparator circuit **41** includes a differential input circuit **71** and an output circuit **72**. The differential input circuit **71** includes a mirror pair **81**, a differential pair **82**, a constant-current supply **83**, and a switching circuit **84**. The mirror pair **81** has active elements **81p** and **81m**. The differential pair **82** has active elements **82p** and **82m**. The active elements **81p**, **81m**, **82p**, and **82m** are metal oxide semiconductor field-effect transistors (MOSFETs). The active elements **81p**, **81m**, **82p**, and **82m** may be active elements other than MOSFETs.

[0148] The mirror pair **81** feeds a reference current through one of the active elements **81p** and **81m** and feeds a current that is a duplicate of the reference current through the other of the active elements **81p** and **81m**.

[0149] The differential pair **82** feeds, through the active element **82p**, a current corresponding to the first input voltage **Vin1** input to the first input terminal **41a** of the comparator circuit **41** and feeds, through the active element **82m**, a current corresponding to the second input voltage **Vin2** input to the second input terminal **41b** of the comparator circuit **41**. The current fed through the active element **82p** includes the current fed through the active element **81p**. The current fed through the active element **82m** includes the current fed through the active element **81m**.

[0150] The constant-current supply **83** feeds a constant current. The constant current fed by the constant-current supply **83** is a current that is a merge of the current fed through the active element **82p** and the current fed through the active element **82m**.

[0151] The switching circuit **84** receives the control signal **S3**. Upon receiving the control signal **S3** being the second control signal Low, the switching circuit **84** brings the active element **81m** into diode connection to feed the reference current through the active element **81m** and feeds the duplicate of the reference current through the active element **81p**, as illustrated in FIG. **6A**. Upon receiving the control signal **S3** being the second control signal Low, the switching circuit **84** also connects the output circuit **72** electrically to the active element **82p** to feed, through the active element **82p**, a current that is a merge of the current flowing into the active element **81p** and current flowing out of the output circuit **72**. Upon receiving the control signal **S3** being the first control signal High, the switching circuit **84** brings the active element **81p** into diode connection to feed the reference current through the active element **81p** and feeds the duplicate of the reference current through the active element

81m, as illustrated in FIG. 6B. Upon receiving the control signal **S3** being the first control signal High, the switching circuit **84** also connects the output circuit **72** electrically to the active element **82m** to feed, through the active element **82m**, a current that is a merge of the current flowing into the active element **81m** and the current flowing out of the output circuit **72**.

[0152] The output circuit **72** outputs the output voltage **Vout** corresponding to an outgoing current, from the output terminal **41c** of the comparator circuit **41**.

[0153] As described above, the switching circuit **84** sets the first input terminal **41a** and second input terminal **41b** of the comparator circuit **41** as a non-inverting input terminal and an inverting input terminal, respectively, while the control signal **S3** is the second control signal Low, and the switching circuit **84** sets the first input terminal **41a** and second input terminal **41b** of the comparator circuit **41** as an inverting input terminal and a non-inverting input terminal, respectively, while the received control signal **S3** is the first control signal High. Thus, while the control signal **S3** is the second control signal Low, the switching circuit **84** sets the output voltage **Vout** at the first output voltage High in response to the first input voltage **Vin1** higher than the second input voltage **Vin2**, and the switching circuit **84** sets the output voltage **Vout** at the second control signal Low in response to the first input voltage **Vin1** lower than the second input voltage **Vin2**. Further, while the control signal **S3** is the first control signal High, the switching circuit **84** sets the output voltage **Vout** at the second output voltage Low in response to the first input voltage **Vin1** higher than the second input voltage **Vin2**, and the switching circuit **84** sets the output voltage **Vout** at the first output voltage High in response to the first input voltage **Vin1** lower than the second input voltage **Vin2**.

2.3 Operation of Temperature Measuring Circuit

[0154] FIGS. 7A and 7B are flowcharts showing the operation of the temperature measuring circuit according to the second embodiment.

[0155] The temperature measuring circuit **2** executes Steps **S121** to **S154** shown in FIGS. 7A and 7B.

[0156] The temperature measuring circuit **2** executes Steps **S121** to **S136** during the first period **T1** to turn alternately into State **1a** illustrated in FIG. 5A and State **2a** illustrated in FIG. 5B, or turn alternately into State **1b** illustrated in FIG. 5C and State **2b** illustrated FIG. 5D. The temperature measuring circuit **2** executes Steps **S138** to **S153** during the subsequent second period **T2** to turn alternately into State **3a** illustrated in FIG. 5E and State **4a** illustrated in FIG. 5F, or turn alternately into State **3b** illustrated in FIG. 5G and State **4b** illustrated FIG. 5H.

[0157] The temperature measuring circuit **2** executes Steps **S122** to **S124** during the period **T1A**, which is included in the first period **T1**, to turn into State **1a** illustrated in FIG. 5A. The temperature measuring circuit **2** executes Steps **S125** to **S127** during the period **T1B**, which is included in the first period **T1**, to turn into State **2a** illustrated in FIG. 5B. The temperature measuring circuit **2** executes Steps **S130** to **S132** during a period **T1C**, which is included in the first period **T1**, to turn into State **1b** illustrated in FIG. 5C. The temperature measuring circuit **2** executes Steps **S133** to **S135** during a period **T1D**, which is included in the first period **T1**, to turn into State **2b** illustrated in FIG. 5D.

[0158] The temperature measuring circuit **2** executes Steps **S139** to **S141** during the period **T2A**, which is included in the second period **T2**, to turn into State **3a** illustrated in FIG. 5E. The temperature measuring circuit **2** executes Steps **S142** to **S144** during the period **T2B**, which is included in the second period **T2**, to turn into State **4a** illustrated in FIG. 5F. The temperature measuring circuit **2** executes Steps **S147** to **S149** during a period **T2C**, which is included in the second period **T2**, to turn into State **3b** illustrated in FIG. 5G. The temperature measuring circuit **2** executes Steps **S150** to **S152** during a period **T2D**, which is included in the second period **T2**, to turn into State **4b** illustrated in FIG. 5H.

Initialization of Measurement Number

[0159] In Step **S121**, the switching control circuit **53** initializes a measurement number **i** to zero. The measurement number **i** denotes how many times the control signal **S1** has been set at the first control signal High.

Charge During Period T1A

[0160] In subsequent Step **S122**, as illustrated in FIG. 5A, the control circuit **42** sets the control signal **S1** at the first control signal High and sets the control signal **S2** at the first control signal High. In addition, the switching control circuit **53** sets the control signal **S3** at the second control signal Low. The switching control circuit **53** also increments the measurement number **i**.

[0161] Accordingly, the positive electrode **21b** of the first current supply **21** is electrically connected to the terminal **11a** of the capacitor **11**. Further, the first input terminal **41a** of the comparator circuit **41** is electrically connected to the positive electrode **31b** of the first voltage supply **31**. Further, the second input terminal **41b** of the comparator circuit **41** is electrically connected to the terminal **11a** of the capacitor **11**. Further, the first input terminal **41a** and second input terminal **41b** of the comparator circuit **41** are set as a non-inverting input terminal and an inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit **41** is electrically connected to the positive electrode **31b** of the first voltage supply **31**. In addition, the inverting input terminal of the comparator circuit **41** is electrically connected to the terminal **11a** of the capacitor **11**.

[0162] Accordingly, the first current supply **21** charges the capacitor **11**. Further, the comparator circuit **41** compares the first reference voltage **Vref** and the inter-terminal voltage **Vc** and sets the output voltage **Vout** at the first output voltage High.

[0163] In subsequent Step **S123**, the control circuit **42** determines whether the time **TM1** elapsed from the start of the first period **T1** is shorter than the first measurement time **TF1**. If determining that the time **TM1** is shorter than the first measurement time **TF1**, the control circuit **42** executes Step **S124**. If determining that the time **TM1** is equal to or longer than the first measurement time **TF1**, the control circuit **42** executes Step **S137**.

[0164] In Step **S124**, the control circuit **42** determines whether the inter-terminal voltage **Vc** is lower than the first reference voltage **Vref** on the basis of the output voltage **Vout**. If determining that the inter-terminal voltage **Vc** is lower than the first reference voltage **Vref**, the control circuit **42** executes Step **S122**. If determining that the inter-terminal

voltage V_c is equal to or higher than the first reference voltage V_{ref} , the control circuit 42 executes Step S125.

[0165] From Steps S122 to S124, the control circuit 42 causes the first current supply 21 to continue charging the capacitor 11 and causes the comparator circuit 41 to continue comparing the first reference voltage V_{ref} and the inter-terminal voltage V_c until the time $TM1$ reaches the first measurement time $TF1$, or until the inter-terminal voltage V_c rises to the first reference voltage V_{ref} . The control circuit 42 ends the first period T1 and starts the second period T2 in synchronization with that the time $TM1$ reaches the first measurement time $TF1$. The control circuit 42 ends the period T1A and starts the period T1B in synchronization with that the inter-terminal voltage V_c rises to the first reference voltage V_{ref} , and that the output voltage V_{out} changes from the first output voltage High to the second output voltage Low.

Discharge During Period T1B

[0166] In Step S125, as illustrated in FIG. 5B, the control circuit 42 sets the control signal S1 at the second control signal Low and sets the control signal S2 at the first control signal High. The control circuit 42 may set the control signal S2 at the second control signal Low. In addition, the switching control circuit 53 sets the control signal S3 at the second control signal Low.

[0167] Accordingly, the negative electrode 22a of the second current supply 22 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the positive electrode 32b of the second voltage supply 32. Further, the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are set as a non-inverting input terminal and an inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. In addition, the inverting input terminal of the comparator circuit 41 is electrically connected to the positive electrode 32b of the second voltage supply 32.

[0168] Accordingly, the second current supply 22 discharges the capacitor 11. Further, the comparator circuit 41 compares the inter-terminal voltage V_c and the second reference voltage V_{ref}/M and sets the output voltage V_{out} at the first output voltage High.

[0169] In subsequent Step S126, the control circuit 42 determines whether the time $TM1$ is shorter than the first measurement time $TF1$. If determining that the time $TM1$ is shorter than the first measurement time $TF1$, the control circuit 42 executes Step S127. If determining that the time $TM1$ is equal to or longer than the first measurement time $TF1$, the control circuit 42 executes Step S137.

[0170] In Step S127, the control circuit 42 determines whether the inter-terminal voltage V_c is higher than the second reference voltage V_{ref}/M on the basis of the output voltage V_{out} . If determining that the inter-terminal voltage V_c is higher than the second reference voltage V_{ref}/M , the control circuit 42 executes Step S125. If determining that the inter-terminal voltage V_c is equal to or lower than the second reference voltage V_{ref}/M , the control circuit 42 executes Step S128.

[0171] From Steps S125 to S127, the control circuit 42 causes the second current supply 22 to continue discharging the capacitor 11 and causes the comparator circuit 41 to continue comparing the inter-terminal voltage V_c and the second reference voltage V_{ref}/M until the time $TM1$ reaches the first measurement time $TF1$, or until the inter-terminal voltage V_c lowers to the second reference voltage V_{ref}/M . The control circuit 42 ends the first period T1 and starts the second period T2 in synchronization with that the time $TM1$ reaches the first measurement time $TF1$. The control circuit 42 ends the period T1B and starts the period T1A or TIC in synchronization with that the inter-terminal voltage V_c lowers to the second reference voltage V_{ref}/M , and that the output voltage V_{out} changes from the first output voltage High to the second output voltage Low.

Determination of Measurement Number

[0172] In Step S128, the switching control circuit 53 determines whether the measurement number i has reached the setpoint N . If determining that the measurement number i has reached the setpoint N , the switching control circuit 53 executes Step S129. If determining that the measurement number i has not yet reached the setpoint N , the switching control circuit 53 executes Step S122.

[0173] From Steps S121 to S128, until the measurement number i reaches the setpoint N , the control circuit 42 alternately performs the following two kinds of operations: one is causing the first current supply 21 to charge the capacitor 11, and causing the comparator circuit 41 to compare the inter-terminal voltage V_c and the first reference voltage V_{ref} , and the other is causing the second current supply 22 to discharge the capacitor 11, and causing the comparator circuit 41 to compare the inter-terminal voltage V_c and the second reference voltage V_{ref}/M . The control circuit 42 ends the period T1B and starts the period TIC in synchronization with that the measurement number i reaches the setpoint N .

Initialization of Measurement Number

[0174] In Step S129, the switching control circuit 53 initializes the measurement number i to zero.

Charge During Period TIC

[0175] In subsequent Step S130, as illustrated in FIG. 5C, the control circuit 42 sets the control signal S1 at the first control signal High and sets the control signal S2 at the first control signal High. In addition, the switching control circuit 53 sets the control signal S3 at the first control signal High. The switching control circuit 53 also increments the measurement number i .

[0176] Accordingly, the positive electrode 21b of the first current supply 21 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the positive electrode 31b of the first voltage supply 31. Further, the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are set as an inverting input terminal and a non-inverting input terminal, respectively. Accordingly, like in Step S122, the non-inverting input terminal of the comparator circuit 41 is electrically connected to the positive electrode 31b of the first voltage

supply 31. In addition, the inverting input terminal of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11.

[0177] Accordingly, the first current supply 21 charges the capacitor 11. Further, the comparator circuit 41 compares the first reference voltage Vref and the inter-terminal voltage Vc and sets the output voltage Vout at the first output voltage High.

[0178] In subsequent Step S131, the control circuit 42 determines whether the time TM1 elapsed from the start of the first period T1 is shorter than the first measurement time TF1. If determining that the time TM1 is shorter than the first measurement time TF1, the control circuit 42 executes Step S132. If determining that the time TM1 is equal to or longer than the first measurement time TF1, the control circuit 42 executes Step S137.

[0179] In Step S132, the control circuit 42 determines whether the inter-terminal voltage Vc is lower than the first reference voltage Vref on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is lower than the first reference voltage Vref, the control circuit 42 executes Step S130. If determining that the inter-terminal voltage Vc is equal to or higher than the first reference voltage Vref, the control circuit 42 executes Step S133.

[0180] From Steps S130 to S132, the control circuit 42 causes the first current supply 21 to continue charging the capacitor 11 and causes the comparator circuit 41 to continue comparing the first reference voltage Vref and the inter-terminal voltage Vc until the time TM1 reaches the first measurement time TF1, or until the inter-terminal voltage Vc rises to the first reference voltage Vref. The control circuit 42 ends the first period T1 and starts the second period T2 in synchronization with that the time TM1 reaches the first measurement time TF1. The control circuit 42 ends the period TIC and starts the period T1D in synchronization with that the inter-terminal voltage Vc rises to the first reference voltage Vref, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Discharge During Period T1D

[0181] In Step S133, the control circuit 42 sets the control signal S1 at the second control signal Low and sets the control signal S2 at the first control signal High. The control circuit 42 may set the control signal S2 at the second control signal Low. In addition, the switching control circuit 53 sets the control signal S3 at the first control signal High.

[0182] Accordingly, the negative electrode 22a of the second current supply 22 is electrically connected to the terminal 11a of the capacitor 11, as illustrated in FIG. 5D. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the positive electrode 32b of the second voltage supply 32. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are set as an inverting input terminal and a non-inverting input terminal, respectively. Accordingly, like in Step S125, the non-inverting input terminal of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. In addition, the inverting input terminal of the comparator circuit 41 is electrically connected to the positive electrode 32b of the second voltage supply 32.

[0183] Accordingly, the second current supply 22 discharges the capacitor 11. Further, the comparator circuit 41 compares the inter-terminal voltage Vc and the second reference voltage Vref/M and sets the output voltage Vout at the first output voltage High.

[0184] In subsequent Step S134, the control circuit 42 determines whether the time TM1 is shorter than the first measurement time TF1. If determining that the time TM1 is shorter than the first measurement time TF1, the control circuit 42 executes Step S135. If determining that the time TM1 is equal to or longer than the first measurement time TF1, the control circuit 42 executes Step S137.

[0185] In Step S135, the control circuit 42 determines whether the inter-terminal voltage Vc is higher than the second reference voltage Vref/M on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is higher than the second reference voltage Vref/M, the control circuit 42 executes Step S133. If determining that the inter-terminal voltage Vc is equal to or lower than the second reference voltage Vref/M, the control circuit 42 executes Step S136.

[0186] From Steps S133 to S135, the control circuit 42 causes the second current supply 22 to continue discharging the capacitor 11 and causes the comparator circuit 41 to continue comparing the inter-terminal voltage Vc and the second reference voltage Vref/M until the time TM1 reaches the first measurement time TF1, or until the inter-terminal voltage Vc lowers to the second reference voltage Vref/M. The control circuit 42 ends the first period T1 and starts the second period T2 in synchronization with that the time TM1 reaches the first measurement time TF1. The control circuit 42 ends the period T1D and starts the period TIC in synchronization with that the inter-terminal voltage Vc lowers to the second reference voltage Vref/M, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Determination of Measurement Number

[0187] In Step S136, the switching control circuit 53 determines whether the measurement number i has reached the setpoint N. If determining that the measurement number i has reached the setpoint N, the switching control circuit 53 executes Step S121. If determining that the measurement number i has not yet reached the setpoint N, the switching control circuit 53 executes Step S130.

[0188] From Steps S129 to S136, until the measurement number i reaches the setpoint N, the control circuit 42 alternately performs the following two kinds of operations: one is causing the first current supply 21 to charge the capacitor 11, and causing the comparator circuit 41 to compare the inter-terminal voltage Vc and the first reference voltage Vref, and the other is causing the second current supply 22 to discharge the capacitor 11, and causing the comparator circuit 41 to compare the inter-terminal voltage Vc and the second reference voltage Vref/M. The control circuit 42 ends the period T1D and starts the period T1A in synchronization with that the measurement number i reaches the setpoint N.

[0189] From Steps S121 to S136, during the first period T1, the non-inverting input terminal and inverting input terminal of the comparator circuit 41 are switched, and in synchronization with their switching, the voltages input to the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are switched. This can cancel the

effect of an input offset resulting from variations in an active element, such as a transistor, provided in the comparator circuit 41.

Reset of Counter Circuit and Change of Count Writing Destination

[0190] In Step S137, the counter circuit 43 is reset, and the destination into which the foregoing number of periodic changes counted by the counter circuit 43 is written is changed from the first address to the second address.

Initialization of Measurement Number

[0191] In Step S138, the switching control circuit 53 initializes the measurement number i to zero.

Charge During Period T2A

[0192] In Step S139, as illustrated in FIG. 5E, the control circuit 42 sets the control signal S1 at the first control signal High and sets the control signal S2 at the second control signal Low. In addition, the switching control circuit 53 sets the control signal S3 at the second control signal Low. The switching control circuit 53 also increments the measurement number i .

[0193] Accordingly, the positive electrode 21b of the first current supply 21 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the positive electrode 33b of the third voltage supply 33. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are set as a non-inverting input terminal and an inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit 41 is electrically connected to the positive electrode 33b of the third voltage supply 33. In addition, the inverting input terminal of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11.

[0194] Accordingly, the first current supply 21 charges the capacitor 11. Further, the comparator circuit 41 compares the third reference voltage Vbe and the inter-terminal voltage Vc and sets the output voltage Vout at the first output voltage High.

[0195] In subsequent Step S140, the control circuit 42 determines whether the time TM2 elapsed from the start of the second period T2 is shorter than the second measurement time TF2. If determining that the time TM2 is shorter than the second measurement time TF2, the control circuit 42 executes Step S141. If determining that the time TM2 is equal to or longer than the second measurement time TF2, the control circuit 42 executes Step S154.

[0196] In Step S141, the control circuit 42 determines whether the inter-terminal voltage Vc is lower than the third reference voltage Vbe on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is lower than the third reference voltage Vbe, the control circuit 42 executes Step S139. If determining that the inter-terminal voltage Vc is equal to or higher than the third reference voltage Vbe, the control circuit 42 executes Step S142.

[0197] From Steps S139 to S141, the control circuit 42 causes the first current supply 21 to continue charging the

capacitor 11 and causes the comparator circuit 41 to continue comparing the third reference voltage Vbe and the inter-terminal voltage Vc until the time TM2 reaches the second measurement time TF2, or until the inter-terminal voltage Vc rises to the third reference voltage Vbe. The control circuit 42 ends the second period T2 in synchronization with that the time TM2 reaches the second measurement time TF2. The control circuit 42 ends the period T2A and starts the period T2B in synchronization with that the inter-terminal voltage Vc rises to the third reference voltage Vbe, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Discharge During Period T2B

[0198] In Step S142, the control circuit 42 sets the control signal S1 at the second control signal Low and sets the control signal S2 at the second control signal Low. The control circuit 42 may set the control signal S2 at the first control signal High. In addition, the switching control circuit 53 sets the control signal S3 at the second control signal Low.

[0199] Accordingly, the negative electrode 22a of the second current supply 22 is electrically connected to the terminal 11a of the capacitor 11, as illustrated in FIG. 5F. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the positive electrode 32b of the second voltage supply 32. Further, the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are set as a non-inverting input terminal and an inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. In addition, the inverting input terminal of the comparator circuit 41 is electrically connected to the positive electrode 32b of the second voltage supply 32.

[0200] Accordingly, the second current supply 22 discharges the capacitor 11. Further, the comparator circuit 41 compares the inter-terminal voltage Vc and the second reference voltage Vref/M and sets the output voltage Vout at the first output voltage High.

[0201] In subsequent Step S143, the control circuit 42 determines whether the time TM2 is shorter than the second measurement time TF2. If determining that the time TM2 is shorter than the second measurement time TF2, the control circuit 42 executes Step S144. If determining that the time TM2 is equal to or longer than the second measurement time TF2, the control circuit 42 executes Step S154.

[0202] In Step S144, the control circuit 42 determines whether the inter-terminal voltage Vc is higher than the second reference voltage Vref/M on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is higher than the second reference voltage Vref/M, the control circuit 42 executes Step S142. If determining that the inter-terminal voltage Vc is equal to or lower than the second reference voltage Vref/M, the control circuit 42 executes Step S145.

[0203] From Steps S142 to S144, the control circuit 42 causes the second current supply 22 to continue discharging the capacitor 11 and causes the comparator circuit 41 to continue comparing the inter-terminal voltage Vc and the second reference voltage Vref/M until the time TM2 reaches

the second measurement time TF2, or until the inter-terminal voltage Vc lowers to the second reference voltage Vref/M. The control circuit 42 ends the second period T2 in synchronization with that the time TM2 reaches the second measurement time TF2. The control circuit 42 ends the period T2B and starts the period T2A or T2C in synchronization with that the inter-terminal voltage Vc lowers to the second reference voltage Vref/M, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Determination of Measurement Number

[0204] In Step S145, the switching control circuit 53 determines whether the measurement number i has reached the setpoint N. If determining that the measurement number i has reached the setpoint N, the switching control circuit 53 executes Step S146. If determining that the measurement number i has not yet reached the setpoint N, the switching control circuit 53 executes Step S139.

[0205] From Steps S138 to S145, until the measurement number i reaches the setpoint N, the control circuit 42 alternately performs the following two kinds of operations: one is causing the first current supply 21 to charge the capacitor 11, and causing the comparator circuit 41 to compare the inter-terminal voltage Vc and the third reference voltage Vbe; and the other is causing the second current supply 22 to discharge the capacitor 11, and causing the comparator circuit 41 to compare the inter-terminal voltage Vc and the second reference voltage Vref/M. The control circuit 42 ends the period T2B and starts the period T2C in synchronization with that the measurement number i reaches the setpoint N.

Initialization of Measurement Number

[0206] In Step S146, the switching control circuit 53 initializes the measurement number i to zero.

Charge During Period T2C

[0207] In Step S147, as illustrated in FIG. 5G, the control circuit 42 sets the control signal S1 at the first control signal High and sets the control signal S2 at the second control signal Low. In addition, the switching control circuit 53 sets the control signal S3 at the first control signal High. The switching control circuit 53 also increments the measurement number i.

[0208] Accordingly, the positive electrode 21b of the first current supply 21 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the positive electrode 33b of the second voltage supply 33. Further, the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are set as an inverting input terminal and a non-inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit 41 is electrically connected to the positive electrode 33b of the third voltage supply 33. In addition, the inverting input terminal of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11.

[0209] Accordingly, the first current supply 21 charges the capacitor 11. Further, the comparator circuit 41 compares the

third reference voltage Vbe and the inter-terminal voltage Vc and sets the output voltage Vout at the first output voltage High.

[0210] In subsequent Step S148, the control circuit 42 determines whether the time TM2 elapsed from the start of the second period T2 is shorter than the second measurement time TF2. If determining that the time TM2 is shorter than the second measurement time TF2, the control circuit 42 executes Step S149. If determining that the time TM2 is equal to or longer than the second measurement time TF2, the control circuit 42 executes Step S154.

[0211] In Step S149, the control circuit 42 determines whether the inter-terminal voltage Vc is lower than the third reference voltage Vbe on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is lower than the third reference voltage Vbe, the control circuit 42 executes Step S147. If determining that the inter-terminal voltage Vc is equal to or higher than the third reference voltage Vbe, the control circuit 42 executes Step S150.

[0212] From Steps S147 to S149, the control circuit 42 causes the first current supply 21 to continue charging the capacitor 11 and causes the comparator circuit 41 to continue comparing the third reference voltage Vbe and the inter-terminal voltage Vc until the time TM2 reaches the second measurement time TF2, or until the inter-terminal voltage Vc rises to the third reference voltage Vbe. The control circuit 42 ends the second period T2 in synchronization with that the time TM2 reaches the second measurement time TF2. The control circuit 42 ends the period T2C and starts the period T2D in synchronization with that the inter-terminal voltage Vc rises to the third reference voltage Vbe, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Discharge During Period T2D

[0213] In Step S150, as illustrated in FIG. 5H, the control circuit 42 sets the control signal S1 at the second control signal Low and sets the control signal S2 at the second control signal Low. The control circuit 42 may set the control signal S2 at the first control signal High. In addition, the switching control circuit 53 sets the control signal S3 at the first control signal High.

[0214] Accordingly, the negative electrode 22a of the second current supply 22 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a of the comparator circuit 41 is electrically connected to the positive electrode 32b of the second voltage supply 32. Further, the second input terminal 41b of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. Further, the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are set as an inverting input terminal and a non-inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit 41 is electrically connected to the terminal 11a of the capacitor 11. In addition, the inverting input terminal of the comparator circuit 41 is electrically connected to the positive electrode 32b of the second voltage supply 32.

[0215] Accordingly, the second current supply 22 discharges the capacitor 11. Further, the comparator circuit 41 compares the inter-terminal voltage Vc and the second reference voltage Vref/M and sets the output voltage Vout at the first output voltage High.

[0216] In subsequent Step S151, the control circuit 42 determines whether the time TM2 is shorter than the second measurement time TF2. If determining that the time TM2 is shorter than the second measurement time TF2, the control circuit 42 executes Step S152. If determining that the time TM2 is equal to or longer than the second measurement time TF2, the control circuit 42 executes Step S154.

[0217] In Step S152, the control circuit 42 determines whether the inter-terminal voltage V_c is higher than the second reference voltage V_{ref}/M on the basis of the output voltage V_{out} . If determining that the inter-terminal voltage V_c is higher than the second reference voltage V_{ref}/M , the control circuit 42 executes Step S150. If determining that the inter-terminal voltage V_c is equal to or lower than the second reference voltage V_{ref}/M , the control circuit 42 executes Step S153.

[0218] From Steps S150 to S152, the control circuit 42 causes the second current supply 22 to continue discharging the capacitor 11 and causes the comparator circuit 41 to continue comparing the inter-terminal voltage V_c and the second reference voltage V_{ref}/M until the time TM2 reaches the second measurement time TF2, or until the inter-terminal voltage V_c lowers to the second reference voltage V_{ref}/M . The control circuit 42 ends the second period T2 in synchronization with that the time TM2 reaches the second measurement time TF2. The control circuit 42 ends the period T2D and starts the period T2A or T2C in synchronization with that the inter-terminal voltage V_c lowers to the second reference voltage V_{ref}/M , and that the output voltage V_{out} changes from the first output voltage High to the second output voltage Low.

Determination of Measurement Number

[0219] In Step S153, the switching control circuit 53 determines whether the measurement number i has reached the setpoint N . If determining that the measurement number i has reached the setpoint N , the switching control circuit 53 executes Step S138. If determining that the measurement number i has not yet reached the setpoint N , the switching control circuit 53 executes Step S147.

[0220] From Steps S146 to S153, until the measurement number i reaches the setpoint N , the control circuit 42 alternately preforms the following two kinds of operations: one is causing the first current supply 21 to charge the capacitor 11, and causing the comparator circuit 41 to compare the inter-terminal voltage V_c and the third reference voltage V_{be} ; and the other is causing the second current supply 22 to discharge the capacitor 11, and causing the comparator circuit 41 to compare the inter-terminal voltage V_c and the second reference voltage V_{ref}/M .

Counts Readout

[0221] In Step S154, the counter circuit 43 reads the counts from the first and second addresses, and from the counts read from the first and second addresses, the counter circuit 43 obtains the first frequency $F1$ of the control signal S1 during the first period T1, and the second frequency $F2$ of the control signal S1 during the second period T2. Further, the temperature obtaining unit 45 obtains the temperature on the basis of the first frequency $F1$ and second frequency $F2$.

2.4 Time Variations of First Input Voltage, Second Input Voltage, Inter-Terminal Voltage, and Control Signal

Time Variations During First Period T1

[0222] FIG. 8A is a timing chart showing the following time variations during the first period T1: time variations of the first input voltage V_{in1} and the second input voltage V_{in2} , which are input to the comparator provided in the temperature measuring circuit according to the second embodiment; time variations of the inter-terminal voltage V_c , which is generated by the capacitor provided in the temperature measuring circuit; time variations of the control signal S1, which is output by the control circuit provided in the temperature measuring circuit; and time variations of the control signal S3, which is output by the switching control circuit provided in the temperature measuring circuit.

[0223] As shown in FIG. 8A, in the temperature measuring circuit 2 in Steps S121 to S136, the periods T1A and T1B come alternately while the control signal S3 during the first period T1 is at the second control signal Low, i.e., while the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are a non-inverting input terminal and an inverting input terminal, respectively; moreover, the period TIC and T1D come alternately while the control signal S3 during the first period T1 is at the first control signal High, i.e., while the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are an inverting input terminal and a non-inverting input terminal, respectively.

[0224] During the period T1A, the first input voltage V_{in1} and the second input voltage V_{in2} are at the first reference voltage V_{ref} and the inter-terminal voltage V_c , respectively. In addition, the inter-terminal voltage V_c rises along with time. Accordingly, the inter-terminal voltage V_c rises from the second reference voltage V_{ref}/M to the first reference voltage V_{ref} . In addition, the control signal S1 is kept at the first control signal High.

[0225] During the period T1B, the first input voltage V_{in1} and the second input voltage V_{in2} are at the inter-terminal voltage V_c and the second reference voltage V_{ref}/M , respectively. In addition, the inter-terminal voltage V_c lowers along with time. Accordingly, the inter-terminal voltage V_c lowers from the first reference voltage V_{ref} to the second reference voltage V_{ref}/M . In addition, the control signal S1 is kept at the second control signal Low.

[0226] During the period TIC, conversely to the period T1A, the first input voltage V_{in1} and the second input voltage V_{in2} are at the inter-terminal voltage V_c and the first reference voltage V_{ref} , respectively. In addition, the inter-terminal voltage V_c rises along with time. Accordingly, the inter-terminal voltage V_c rises from the second reference voltage V_{ref}/M to the first reference voltage V_{ref} . In addition, the control signal S1 is kept at the first control signal High.

[0227] During the period T1D, conversely to the period T1B, the first input voltage V_{in1} and the second input voltage V_{in2} are at the second reference voltage V_{ref}/M and the inter-terminal voltage V_c , respectively. In addition, the inter-terminal voltage V_c lowers along with time. Accordingly, the inter-terminal voltage V_c lowers from the first reference voltage V_{ref} to the second reference voltage V_{ref}/M . In addition, the control signal S1 is kept at the second control signal Low.

[0228] During the periods T1A and T1C, the non-inverting input terminal and inverting input terminal of the comparator circuit 41 are switched. During the periods T1B and T1D, the non-inverting input terminal and inverting input terminal of the comparator circuit 41 are switched. This can cancel the effect of an input offset resulting from variations in an active element, such as a transistor, provided in the comparator circuit 41.

Time Variations During Second Period T2

[0229] FIG. 8B is a timing chart showing the following time variations during the second period T2: time variations of the first input voltage Vin1 and the second input voltage Vin2, which are input to the comparator provided in the temperature measuring circuit according to the second embodiment; time variations of the inter-terminal voltage Vc, which is generated by the capacitor provided in the temperature measuring circuit; time variations of the control signal S1, which is output by the control circuit provided in the temperature measuring circuit; and time variations of the control signal S3, which is output by the switching control circuit provided in the temperature measuring circuit.

[0230] As shown in FIG. 8B, in the temperature measuring circuit 2 in Steps S138 to S153, the periods T2A and T2B come alternately while the control signal S3 during the second period T2 is at the second control signal Low, i.e., while the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are a non-inverting input terminal and an inverting input terminal, respectively; moreover, the period T2C and T2D come alternately while the control signal S3 during the second period T2 is at the first control signal High, i.e., while the first input terminal 41a and second input terminal 41b of the comparator circuit 41 are an inverting input terminal and a non-inverting input terminal, respectively.

[0231] During the period T2A, the first input voltage Vin1 and the second input voltage Vin2 are at the third reference voltage Vbe and the inter-terminal voltage Vc, respectively. In addition, the inter-terminal voltage Vc rises along with time. Accordingly, the inter-terminal voltage Vc rises from the second reference voltage Vref/M to the third reference voltage Vbe. In addition, the control signal S1 is kept at the first control signal High.

[0232] During the period T2B, the first input voltage Vin1 and the second input voltage Vin2 are at the inter-terminal voltage Vc and the second reference voltage Vref/M, respectively. In addition, the inter-terminal voltage Vc lowers along with time. Accordingly, the inter-terminal voltage Vc lowers from the third reference voltage Vbe to the second reference voltage Vref/M. In addition, the control signal S1 is kept at the second control signal Low.

[0233] During the period T2C, conversely to the period T2A, the first input voltage Vin1 and the second input voltage Vin2 are at the inter-terminal voltage Vc and the third reference voltage Vbe, respectively. In addition, the inter-terminal voltage Vc rises along with time. Accordingly, the inter-terminal voltage Vc rises from the second reference voltage Vref/M to the third reference voltage Vbe. In addition, the control signal S1 is kept at the first control signal High.

[0234] During the period T2D, conversely to the period T2B, the first input voltage Vin1 and the second input voltage Vin2 are at the second reference voltage Vref/M and the inter-terminal voltage Vc, respectively. In addition, the

inter-terminal voltage Vc lowers along with time. Accordingly, the inter-terminal voltage Vc lowers from the third reference voltage Vbe to the second reference voltage Vref/M. In addition, the control signal S1 is kept at the second control signal Low.

[0235] During the periods T2A and T2C, the non-inverting input terminal and inverting input terminal of the comparator circuit 41 are switched. During the periods T2B and T2D, the non-inverting input terminal and inverting input terminal of the comparator circuit 41 are switched. This can cancel the effect of an input offset resulting from variations in an active element, such as a transistor, provided in the comparator circuit 41.

[0236] The present disclosure is not limited to the above-described embodiment. The present disclosure may be replaced with a configuration substantially identical to those described in the above-described embodiment, a configuration that provides the same action and effect as those described in the above-described embodiment, or a configuration that can achieve the same object as those described in the above-described embodiment.

[0237] While there have been described what are at present considered to be certain embodiments of the invention, it will be understood that various modifications may be made thereto, and it is intended that the appended claim cover all such modifications as fall within the true spirit and scope of the invention.

What is claimed is:

1. A temperature measuring circuit comprising:
 - a capacitor configured to generate an inter-terminal voltage corresponding to an amount of stored electric charge;
 - a first current supply configured to feed a first current;
 - a second current supply configured to feed a second current;
 - a first voltage supply configured to generate a first reference voltage;
 - a second voltage supply configured to generate a second reference voltage;
 - a third voltage supply configured to generate a third reference voltage corresponding to a temperature;
 - a comparator circuit configured to output an output voltage indicating levels of two input voltages that are received;
 - a first switch circuit configured to,
 - upon receiving a first control signal, charge the capacitor through the first current and set the two input voltages at a selected reference voltage and the inter-terminal voltage, and
 - upon receiving a second control signal, discharge the capacitor through the second current and set the two input voltages at the second reference voltage and the inter-terminal voltage;
 - a second switch circuit configured to switch between a first connection state in which the selected reference voltage is set at the first reference voltage, and a second connection state in which the selected reference voltage is set at the third reference voltage;
 - a control circuit configured to,
 - during a first period, set a control signal at the second control signal in response to the output voltage indicating that the inter-terminal voltage is higher than the first reference voltage while the control signal is set at the first control signal and the second

switch circuit is in the first connection state, and set the control signal at the first control signal and bring the second switch circuit into the first connection state in response to the output voltage indicating that the inter-terminal voltage is lower than the second reference voltage while the control signal is set at the second control signal, and

during a second period, set the control signal at the second control signal in response to the output voltage indicating that the inter-terminal voltage is higher than the third reference voltage while the control signal is set at the first control signal and the second switch circuit is in the second connection state, and set the control signal at the first control signal and bring the second switch circuit into the second connection state in response to the output voltage indicating that the inter-terminal voltage is lower than the second reference voltage while the control signal is set at the second control signal;

a counter circuit configured to count a first frequency of the control signal during the first period, and a second frequency of the control signal during the second period; and

a temperature obtaining unit configured to obtain the temperature based on the first and second frequencies.

2. The temperature measuring circuit according to claim 1, wherein the first and second reference voltages are stable with respect to the temperature.

3. The temperature measuring circuit according to claim 1, wherein the second reference voltage is $1/M$ times of the first reference voltage, where M is a positive value greater than 1.

4. The temperature measuring circuit according to claim 1, wherein

the third voltage supply includes a bipolar transistor having a base, a collector, and an emitter, and the third reference voltage is generated between the base and the emitter.

5. The temperature measuring circuit according to claim 1, wherein the temperature obtaining unit determines the third reference voltage from the first and second reference voltages and the first and second frequencies in accordance with following Mathematical Expression (1), and obtains the temperature from a temperature property of the third reference voltage:

[Mathematical Expression 1]

$$V_{be} = \frac{F1}{F2} \times \left(V_{ref} - \frac{V_{ref}}{M} \right) + \frac{V_{ref}}{M}, \quad (1)$$

where V_{ref} denotes the first reference voltage, where V_{ref}/M denotes the second reference voltage, where $F1$ denotes the first frequency, where $F2$ denotes the second frequency, and where V_{be} denotes the third reference voltage.

6. The temperature measuring circuit according to claim 1 comprising a switching control circuit configured to divide the control signal and output the control signal that has been divided.

7. The temperature measuring circuit according to claim 6, wherein

the comparator circuit includes

a first input terminal to which one of the two input voltages is input,

a second input terminal to which another of the two input voltages is input, and

a switching circuit configured to,

while the control signal that has been divided is the first control signal, set the output voltage at a first output voltage in response to the one higher than the another, and set the output voltage at a second output voltage in response to the one lower than the another, and

while the control signal that has been divided is the second control signal, set the output voltage at the second output voltage in response to the one higher than the another, and set the output voltage at the first output voltage in response to the one lower than the another.

8. The temperature measuring circuit according to claim 1, wherein

the two input voltages are a first input voltage and a second input voltage,

the comparator circuit sets the output voltage at a first output voltage in response to the first input voltage higher than the second input voltage, and sets the output voltage at the second output voltage in response to the first input voltage lower than the second input voltage,

setting the selected reference voltage and the inter-terminal voltage at the two input voltages is equivalent to setting the selected reference voltage and the inter-terminal voltage at the first input voltage and the second input voltage, respectively,

setting the second reference voltage and the inter-terminal voltage at the two input voltages is equivalent to setting the second reference voltage and the inter-terminal voltage at the second input voltage and the first input voltage, respectively,

the output voltage indicating that the inter-terminal voltage is higher than the first reference voltage is equivalent to a change of the output voltage from the first reference voltage to the second output voltage,

the output voltage indicating that the inter-terminal voltage is lower than the second reference voltage is equivalent to a change of the output voltage from the first output voltage to the second output voltage, and

the output voltage indicating that the inter-terminal voltage is higher than the third reference voltage is equivalent to a change of the output voltage from the first output voltage to the second output voltage.

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